# HOT SPOTS ON SYMMETRIC CONVEX BODIES OF REVOLUTION

Abstract. We give direct variational proofs of the strict hot-spots property for explicit classes of smooth convex bodies of revolution in every dimension $n \geq 2$. For a body symmetric across its equatorial plane, the first eigenspace consists of the lowest axial-odd mode, the lowest transverse-odd modes, or their direct sum. We prove strict directional monotonicity for each of the two types whenever it belongs to the first eigenspace. Consequently every first eigenfunction has strict hot spots whenever the axial and transverse eigenvalues are distinct. We provide explicit second-moment, long-body, and flat-body conditions which ensure this separation. A pointwise condition on the meridional boundary distance gives further spectral selection, with equality only for the ball when that distance is monotone. For a quartic family sufficiently close to the ball, we prove that there is a unique coincidence aspect ratio and compute its first-order expansion. For three-dimensional circular double cones, we prove a unique coincidence ratio and determine the ordering on either side. The smooth-body hot-spots proofs use a convex tangent-vector variational principle, with its equality case proved in full, and sign changes of the meridional components of a gradient, without assuming an external monotonicity theorem. Smooth positively curved examples, including nonellipsoidal quartic bodies, satisfy $d(\Omega)^n/|\Omega| \to \infty$. Arbitrary mixtures at an axial–transverse eigenvalue coincidence are outside the general hot-spots conclusion of this paper, except for the ball, which is treated separately.

**Draft status.** This candidate proof was obtained by our team on October 18, 2026, through extensive dialogue-based training with AI. The team is currently working hard to verify all the proofs.

Contents

1. Introduction and main results 2
1.1. Research context 2
1.2. The geometric class and the spectral alternatives 2
1.3. Explicit geometric conditions 4
1.4. A pointwise profile criterion 4
1.5. Quartic bodies and spectral transitions 5
1.6. Outline of the proof 5
2. A vector variational principle on a smooth convex domain 6
3. The first eigenspace and the axial variational proof 9
3.1. Odd ground states and spectral classification 10
3.2. Changing the signs of a meridional gradient 11
3.3. A transverse slice barrier 13
4. The transverse first modes and flat bodies 13
4.1. Explicit conditions selecting the transverse branch 17
5. Geometric conditions that select the axial eigenfunction 18
5.1. A lower bound for the transverse ball eigenvalue 18
5.2. The axial second moment 19
5.3. Comparison with the double cone 19
5.4. Completion of the main theorems 21
6. Balls, axial stretching, and examples with unbounded ratio 21
6.1. The ball, including its full multiple eigenspace 21

*Date*: October 18, 2026.

2020 *Mathematics Subject Classification.* Primary 35P05; Secondary 35J25, 35P15.

*Key words and phrases.* Hot spots conjecture, Neumann eigenfunctions, convex domains, bodies of revolution, directional monotonicity, variational methods.

6.2. Only finitely many possible exceptional stretches 22
6.3. An explicit smooth nonellipsoidal family 23
6.4. Exact constants in dimension three 25
7. A profile criterion for spectral ordering 26
7.1. The linear profile: rhombi and double cones 28
7.2. The unique transition for the three-dimensional double cone 30
7.3. Quartic thresholds and localization of coincidences 41
7.4. A unique quartic coincidence near the ball 42
References 44

## 1. Introduction and main results

For a bounded connected smooth domain $\Omega \subset \mathbb{R}^n$, number the Neumann eigenvalues, with multiplicities, by

$$0 = \mu_0(\Omega) < \mu_1(\Omega) \le \mu_2(\Omega) \le \cdots .$$

The first eigenspace in this paper always means the eigenspace $E_1(\Omega)$ for the first *positive* eigenvalue $\mu_1$. All eigenfunctions are real valued. We call the following the strict hot-spots property:

$$\min_{\partial\Omega} u < u(x) < \max_{\partial\Omega} u \qquad (x \in \Omega, \quad 0 \ne u \in E_1(\Omega)). \tag{1.1}$$

A strictly positive derivative in one fixed direction implies this property and excludes every interior critical point.

1.1. **Research context.** Jerison and Nadirashvili proved the hot-spots property for convex planar domains with two axes of symmetry, including multiple first positive eigenvalues [3]. Chen, Li, and Wang proved monotonicity of the least axially odd eigenfunction on suitable rotationally symmetric domains, and obtained hot-spots conclusions for long domains [1]. Their axial-odd eigenvalue need not be the first positive Neumann eigenvalue, so spectral selection is an essential part of a hot-spots argument. Kennedy and Rohleder developed a vector variational approach to higher-dimensional lip domains and related symmetric domains [4]. Mean-gradient arguments for convex symmetric bodies also appear in Klartag [5, Section 4].

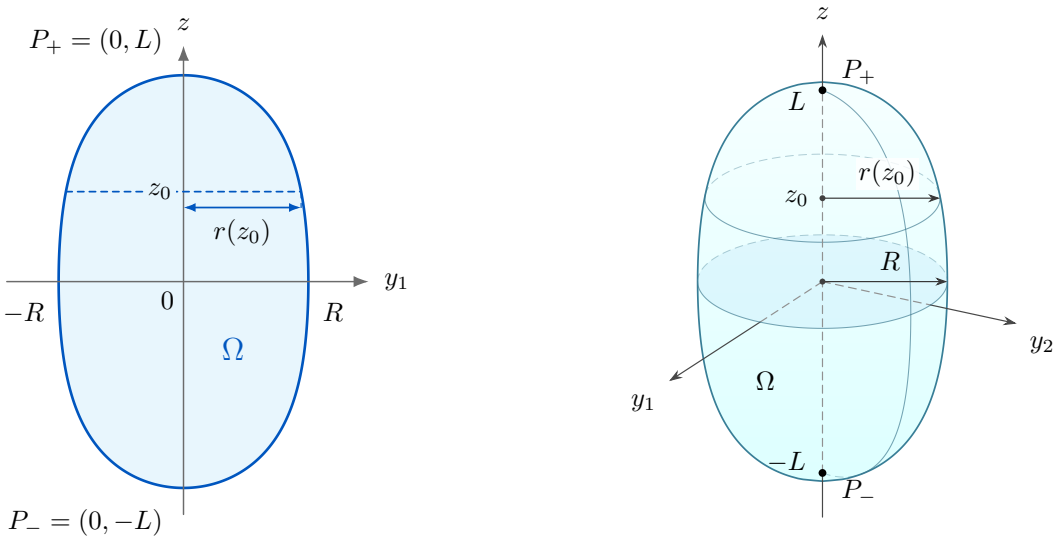
Convexity alone is insufficient in arbitrary dimension: de Dios Pont constructed convex counterexamples in all sufficiently large dimensions [2]. The present argument uses rotational symmetry to reduce the relevant boundary tangency relation to two meridional components. The vector variational inequality and its equality case are proved here, as are the smooth-body monotonicity and spectral-selection steps. For the circular-double-cone transition in Theorem 7.5, we give two proofs of the axial directional estimate. The first uses the axial odd-ground monotonicity theorem of Chen, Li, and Wang [1, Theorem 3.1]; the second derives the needed signs directly from the mixed variational problem on the cone.

1.2. **The geometric class and the spectral alternatives.** Write $x = (y, z) \in \mathbb{R}^{n-1} \times \mathbb{R}$ and $\rho = |y|$. We consider

$$\Omega = \{(y, z) : -L < z < L,\ |y| < r(z)\}, \qquad R = r(0) > 0, \tag{1.2}$$

under the following hypotheses:

(G1) $r$ is even, positive and $C^\infty$ on $(-L, L)$, extends continuously to $[-L, L]$, and $r(\pm L) = 0$;
(G2) $r$ is concave and $r'(z) < 0$ for $0 < z < L$;
(G3) the resulting boundary $\partial\Omega$ is $C^\infty$.




(a) $\Omega \subset \mathbb{R}^2$ (b) $\Omega \subset \mathbb{R}^3$

FIGURE 1. Schematic views of the domain in dimensions two and three, using the same even meridian profile. The section at height $z_0$ has half-width $r(z_0)$ in (a) and is a disk of radius $r(z_0)$ in (b). The shaded equatorial disk in (b) has radius $R = r(0)$; dashed arcs indicate hidden portions of the transparent surface.

These conditions describe smooth convex bodies with strictly decreasing upper meridian profile. Every smooth body of revolution with positive principal curvatures and the stated equatorial symmetry satisfies them. Smoothness is a condition on the hypersurface: the graph $r$ need not be smooth at $\pm L$. In particular, the usual square-root behavior at a curved pole is allowed.

Let $P_\pm = (0, \pm L)$. Figure 1 illustrates the domain and its transverse sections for $n = 2, 3$. Define the two odd-sector eigenvalues by

$$\lambda_\parallel = \inf_{\substack{0 \neq f \in H^1(\Omega) \\ f(y,-z) = -f(y,z)}} \frac{\int_\Omega |\nabla f|^2}{\int_\Omega f^2}, \tag{1.3}$$

$$\lambda_\perp = \inf_{\substack{0 \neq f \in H^1(\Omega) \\ f(-y_1, y_2, \ldots, z) = -f(y_1, y_2, \ldots, z)}} \frac{\int_\Omega |\nabla f|^2}{\int_\Omega f^2}. \tag{1.4}$$

Transverse rotations make the second definition independent of the chosen transverse direction. The symbol $\lambda_{\text{odd}}$ will also denote $\lambda_\parallel$ in the geometric estimates.

**Theorem 1.1** (Spectral alternatives and monotonicity)**.** *For a body satisfying* (G1)–(G3),

$$\mu_1(\Omega) = \min\{\lambda_\parallel, \lambda_\perp\}.$$

*The following conclusions hold.*

(i) *If $\lambda_\parallel < \lambda_\perp$, the first eigenvalue is simple. Its eigenfunction $U$, chosen positive on $z > 0$, is invariant under transverse rotations, odd in $z$, and satisfies*

$$U_z > 0 \quad \text{in } \Omega, \qquad U_\rho < 0 \quad \text{when } z > 0,\ \rho > 0.$$

*Its unique maximum and minimum are $P_+$ and $P_-$.*

(ii) *If $\lambda_\perp < \lambda_\parallel$, the first eigenvalue has multiplicity $n-1$. For every nonzero first eigenfunction, a choice of unit vector $e \in \mathbb{R}^{n-1}$ gives*

$$\partial_e u > 0 \quad \text{in } \Omega.$$

*Its unique maximum and minimum are $(Re, 0)$ and $(-Re, 0)$.*

(iii) *If $\lambda_\parallel = \lambda_\perp$, the multiplicity is $n$. The axial subspace and the transverse subspace separately have the monotonicity and extremum conclusions in* (i) *and* (ii)*. No conclusion for an arbitrary mixture of these two subspaces is included in this assertion.*

*In particular, if $\lambda_\parallel \neq \lambda_\perp$, every first eigenfunction satisfies* (1.1) *and has no interior critical point.*

The distinction at a spectral coincidence is substantive. Monotonicity of separate eigenfunctions does not imply monotonicity of their linear combinations. All geometric hypotheses in the next theorem give a strict spectral separation, so that issue does not arise there.

For a ball, its additional rotational symmetries handle every mixture; Corollary 6.1 gives a proof within the same framework. More generally, fix any profile satisfying (G1)–(G3) and stretch the body in the axial direction by a factor $t > 0$. Proposition 6.2 shows that only finitely many values of $t$ can be spectral coincidences. Thus the full hot-spots conclusion holds throughout every such stretching family apart from a finite set of possible exceptions of precisely this spectral type.

1.3. **Explicit geometric conditions.** Set $\beta_m^2 = \mu_1(B^m)$ for the unit $m$-ball, with $\beta_1 = \pi/2$, and $\alpha_n^2 = \lambda_1^D(B^n)$. Thus $\alpha_n = j_{n/2-1,1}$. Define

$$M_\parallel = \frac{1}{|\Omega|}\int_\Omega z^2 \, \mathrm{d}x, \qquad M_\perp = \frac{1}{|\Omega|}\int_\Omega y_1^2 \, \mathrm{d}x.$$

**Theorem 1.2** (Moment and aspect-ratio conditions)**.** *For a body satisfying* (G1)–(G3)*, either condition*

$$M_\parallel \geq \frac{R^2}{\beta_{n-1}^2} \qquad \text{or} \qquad \frac{L}{R} \geq \frac{\alpha_n}{\beta_{n-1}} \tag{1.5}$$

*implies the axial conclusion of Theorem 1.1. A sufficient condition involving only elementary constants is*

$$\frac{L}{R} \geq \sqrt{\frac{(n+1)(n+2)}{2n}}. \tag{1.6}$$

*In dimensions three and four, $L \geq 2R$ is therefore sufficient.*

*Either condition*

$$M_\perp \geq \frac{4L^2}{\pi^2} \qquad \text{or} \qquad \frac{L}{R} \leq \frac{\pi}{2}\sqrt{\frac{n}{(n+1)(n+2)}} \tag{1.7}$$

*implies the transverse conclusion of Theorem 1.1. All displayed nonstrict inequalities are permitted, including their endpoints.*

1.4. **A pointwise profile criterion.** The profile itself gives another condition selecting the spectral branch. Define

$$\mathcal{F}(z) = z + r(z)r'(z) = \frac{1}{2}\frac{d}{dz}\big(z^2 + r(z)^2\big), \qquad 0 < z < L. \tag{1.8}$$

Thus the sign of $\mathcal{F}$ records whether the distance from the origin to the meridional boundary increases or decreases toward the upper pole.

**Theorem 1.3** (Profile criterion for spectral ordering)**.** *For a body satisfying* (G1)–(G3)*, the following assertions hold.*

(i) *If* $\mathcal{F} \geq 0$ *on* $(0, L)$ *and* $\mathcal{F} \not\equiv 0$*, then* $\lambda_{\|} < \lambda_{\perp}$*.*

(ii) *If* $\mathcal{F} \leq 0$ *on* $(0, L)$ *and* $\mathcal{F} \not\equiv 0$*, then* $\lambda_{\|} > \lambda_{\perp}$*.*

(iii) *If* $\mathcal{F} \equiv 0$*, then* $L = R$ *and* $r(z) = \sqrt{R^2 - z^2}$*, so* $\Omega$ *is a ball and* $\lambda_{\|} = \lambda_{\perp}$*.*

*Consequently, among bodies for which* $z^2 + r(z)^2$ *is monotone on* $[0, L]$*, the two eigenvalues coincide exactly for the ball. Every nonball coincidence requires* $\mathcal{F}$ *to take both positive and negative values on* $(0, L)$*.*

The profile criterion complements the moment conditions in Theorem 1.2. Its proof in Section 7 uses a polarized rotational identity at a spectral coincidence, followed by axial stretching. Corollary 7.2 gives the exact ordering for ellipsoids of revolution. Sign change of $\mathcal{F}$ alone does not determine the ordering or imply coincidence.

1.5. **Quartic bodies and spectral transitions.** The geometric range is also unbounded in the diameter–volume quantity considered here. With $\omega_m = |B^m|$, define

$$\Omega_{L,R,c} = \left\{ (y, z) : \frac{|y|^2}{R^2} + (1 - c)\frac{z^2}{L^2} + c\frac{z^4}{L^4} < 1 \right\}, \qquad 0 < c < 1. \tag{1.9}$$

These are smooth positively curved nonellipsoidal bodies. Section 6 proves that they satisfy the axial theorem for all sufficiently large $L/R$, and that, for $L/R \geq \sqrt{1 + c}$,

$$\frac{d(\Omega_{L,R,c})^n}{|\Omega_{L,R,c}|} = \frac{2^n}{\omega_{n-1} I_{n,c}} \left(\frac{L}{R}\right)^{n-1} \longrightarrow \infty, \qquad I_{n,c} = \int_{-1}^{1} [(1 - s^2)(1 + cs^2)]^{(n-1)/2} \, \mathrm{d}s. \tag{1.10}$$

Here $d(\Omega)$ is the Euclidean diameter. Thus no fixed upper bound on $d(\Omega)^n/|\Omega|$ is required by these results.

For the same quartic family, Corollary 7.8 gives strict axial ordering when $L/R \geq \sqrt{1 + c}$ and strict transverse ordering when $L/R \leq \sqrt{1 - c}$. Every coincidence lies strictly between these thresholds. More precisely, Theorem 7.10 proves that, for each fixed $n \geq 2$ and all sufficiently small $c > 0$, there is exactly one coincidence aspect ratio $a_*(c)$, with

$$a_*(c) = 1 - \frac{n - 2}{2(n + 4)} c + O(c^2).$$

The transverse branch is lower for $L/R < a_*(c)$ and the axial branch is lower for $L/R > a_*(c)$. The function $a_*$ is an exact analytic threshold; the displayed formula is its asymptotic expansion. These nonball coincidences retain the distinction for mixed first eigenfunctions in Theorem 1.1(iii).

1.6. **Outline of the proof.** We first prove the convex tangent-vector inequality $\mathcal{Q}(X) \geq \mu_1 \|X\|_2^2$ and show that equality consists exactly of gradients of first Neumann eigenfunctions. The divergence-free part of a tangent field has zero mean in every Cartesian component; this is the key observation in the proof. Convexity and a patch of positive curvature supplied by an enclosing sphere make the equality case rigid. The resulting injective mean-gradient map reduces the first eigenspace to the two odd-sector alternatives above.

For an axial first eigenfunction, a sign change of its radial derivative and an absolute value of its axial derivative preserve the vector energy and boundary tangency. The equality characterization then forces the desired signs. For a transverse mode, an analogous change lowers an explicit angular energy term, and equality again determines the signs. The constructions are justified in $H^1$ across the axis and equatorial plane; no regularity of polar coordinates at those sets is assumed.

The spectral estimates then follow from transverse slice Poincaré inequalities, linear trial functions, and a comparison with a double-cone weight. Section 6 computes volumes and diameters of the example families and explains why axial stretching produces arbitrarily large diameter–volume ratios within the proved class. Finally, Section 7 derives the pointwise profile criterion from a polarized rotational identity and proves the quartic transition formula by analytic perturbation at the ball. For the circular double cone, a directional-derivative estimate makes every coincidence a strict downward crossing and proves that there is exactly one transition.

## 2. A vector variational principle on a smooth convex domain

Throughout this section, $\Omega \subset \mathbb{R}^n$ is bounded and convex with $C^\infty$ boundary. Write $\mu = \mu_1(\Omega) > 0$ for its first positive Neumann eigenvalue, and set

$$E_1(\Omega) = \{u \in H^1(\Omega) : -\Delta u = \mu u,\ \partial_\nu u = 0\}.$$

The equations in this definition mean that

$$\int_\Omega \nabla u \cdot \nabla \varphi = \mu \int_\Omega u\varphi \qquad \text{for every } \varphi \in H^1(\Omega).$$

Taking $\varphi = 1$ gives zero mean, and elliptic regularity makes these solutions smooth up to the boundary. We use the outward second fundamental form

$$\mathrm{II}_x(S,T) = \langle D_S\nu, T\rangle, \qquad S,T \in T_x\partial\Omega.$$

Thus II is nonnegative on a convex domain. The scalar Poincaré inequality is

$$\int_\Omega |\nabla f|^2 \geq \mu \int_\Omega |f - f_\Omega|^2, \qquad f_\Omega = \frac{1}{|\Omega|}\int_\Omega f. \tag{2.1}$$

For a mean-zero $f$, equality holds precisely when $f \in E_1(\Omega)$, including the zero function. To see the forward implication, take any mean-zero $\varphi \in H^1(\Omega)$ and apply the inequality to $f + t\varphi$. At equality the resulting nonnegative quadratic polynomial has a minimum at $t = 0$, so $\int_\Omega \nabla f \cdot \nabla\varphi = \mu \int_\Omega f\varphi$. Replacing an arbitrary test function by $\varphi - \varphi_\Omega$ extends this identity to all of $H^1(\Omega)$. Conversely, testing the weak eigenvalue equation with $f$ proves equality.

Let

$$\mathcal{X}(\Omega) = \{X \in H^1(\Omega;\mathbb{R}^n) : X \cdot \nu = 0 \text{ on } \partial\Omega \text{ in the trace sense}\}.$$

For $X \in \mathcal{X}(\Omega)$, define

$$\mathcal{Q}(X) = \int_\Omega \left( (\operatorname{div} X)^2 + \sum_{1\leq i<j\leq n} (\partial_i X_j - \partial_j X_i)^2 \right). \tag{2.2}$$

The displayed sum fixes the normalization of the antisymmetric derivative in every dimension.

**Lemma 2.1** (Smooth approximation with tangent boundary values)**.** *The smooth vector fields on $\overline{\Omega}$ which are tangent to $\partial\Omega$ are dense in $\mathcal{X}(\Omega)$ in the $H^1$ norm. Consequently, for every $X \in \mathcal{X}(\Omega)$,*

$$\mathcal{Q}(X) = \int_\Omega |DX|^2 + \int_{\partial\Omega} \mathrm{II}(X,X). \tag{2.3}$$

*Proof.* Take a finite smooth partition of unity on $\overline{\Omega}$ subordinate to interior charts and boundary charts, with each localization supported away from the edges of its chart. On each boundary chart choose a smooth orthonormal frame consisting, on the boundary, of $n-1$ tangent vectors and the normal vector; extend the frame smoothly into a collar. After localization, the normal coefficient of $X$ has zero boundary trace. Its extension by zero outside the chart therefore has zero trace on the whole boundary and belongs to $H^1_0(\Omega)$; the localization vanishes near the lateral edges of the chart, so this extension creates no interior jump. It can be approximated in $H^1$ by smooth coefficients vanishing near the boundary, using the density of $C^\infty_c(\Omega)$ in $H^1_0(\Omega)$. The

tangential coefficients can be approximated by arbitrary smooth coefficients up to the boundary. Multiplication by a further cutoff, equal to one on the support of the localized field, keeps all these approximations inside the chosen chart. Multiplication by this fixed cutoff and by the smooth frame is continuous in $H^1$. Recombining the coefficients therefore produces smooth tangent fields converging to the localized field in $H^1$. Interior smoothing and the partition of unity finish the approximation. The usual scalar approximation up to a smooth boundary used here follows, for example, by a local extension across the flattened boundary and convolution.

For a smooth field, expansion of the squares gives

$$(\operatorname{div} X)^2 + \sum_{i<j}(\partial_i X_j - \partial_j X_i)^2 - |DX|^2 = \operatorname{div}\big(X \operatorname{div} X - (X\cdot\nabla)X\big).$$

On the boundary the first term in this flux vanishes because $X\cdot\nu = 0$. Differentiating that equality in the tangent direction $X$ gives

$$-\langle (X\cdot\nabla)X, \nu\rangle = \langle X, D_X\nu\rangle = \mathrm{II}(X,X).$$

The divergence theorem proves (2.3) for smooth tangent fields. Pass to the just-constructed approximations. The bulk terms are continuous in $H^1$, and the boundary term is continuous under the $L^2$ convergence of traces because II is bounded. This proves the identity for every $X \in \mathcal{X}(\Omega)$. □

**Lemma 2.2** (A patch of positive curvature)**.** *There is a nonempty relatively open patch $\Gamma \subset \partial\Omega$ on which* II *is positive definite. Moreover, if a Neumann eigenfunction vanishes on such a patch, then it vanishes identically. In particular, if $Z \in \mathcal{X}(\Omega)$ has every Cartesian component in $E_1(\Omega)$ and $\int_{\partial\Omega}\mathrm{II}(Z,Z) = 0$, then $Z = 0$.*

*Proof.* Choose $c \in \Omega$, and let $p$ maximize $|x-c|$ on $\overline{\Omega}$. Set $\rho = |p-c| > 0$. The maximum lies on the boundary, since an interior maximum would have $p = c$. Tangential stationarity gives $p - c = a\nu(p)$. The supporting half-space and $c \in \Omega$ give $(c-p)\cdot\nu(p) < 0$, so $a = \rho$. For $T \in T_p\partial\Omega$, choose a smooth boundary curve $\gamma$ with $\gamma(0) = p$ and $\gamma'(0) = T$. Differentiating $\gamma'\cdot\nu(\gamma) = 0$ gives $\gamma''(0)\cdot\nu(p) = -\mathrm{II}_p(T,T)$. The second derivative of $|\gamma - c|^2$ at its maximum is therefore

$$0 \geq 2|T|^2 - 2\rho\,\mathrm{II}_p(T,T).$$

It follows that $\mathrm{II}_p \geq \rho^{-1}I$, and continuity gives the required patch.

For the second assertion, let $-\Delta f = \lambda f$ be a Neumann eigenfunction vanishing on an open boundary patch. Both $f$ and $\partial_\nu f$ are zero there. Choose a ball $B$ centered at a point of the patch with $B\cap\partial\Omega$ contained in that patch, and let $F$ equal $f$ on $B\cap\Omega$ and zero on $B\setminus\overline{\Omega}$. For every $\varphi \in C_c^\infty(B)$, Green's formula gives

$$\int_B F(-\Delta\varphi - \lambda\varphi) = \int_{B\cap\partial\Omega}(\varphi\partial_\nu f - f\partial_\nu\varphi) = 0.$$

There are no terms on $\partial B$ because the test function is compactly supported in $B$. Thus $F$ solves the constant-coefficient Helmholtz equation distributionally throughout $B$. Interior elliptic regularity and analyticity make it a real analytic solution, so its vanishing on the exterior open part forces vanishing throughout $B$. Analytic continuation in the connected interior of $\Omega$ then gives $f = 0$ everywhere. The continuation argument needs only an open smooth boundary patch; its positive curvature is not used in this last step.

Finally, under the hypotheses on $Z$, its components are smooth up to the boundary. The continuous function $\mathrm{II}(Z,Z)$ is nonnegative and has integral zero, so it vanishes on the boundary. Positive definiteness on $\Gamma$ and tangency give $Z = 0$ there. Applying the boundary continuation just proved to each component gives $Z = 0$ throughout $\Omega$. □

**Lemma 2.3** (The tangent-vector ground state)**.** *For every $X \in \mathcal{X}(\Omega)$,*

$$\mathcal{Q}(X) \geq \mu \int_\Omega |X|^2. \tag{2.4}$$

*Equality holds if and only if $X = \nabla p$ for some $p \in E_1(\Omega)$.*

*Proof.* The zero normal trace gives $\int_\Omega \operatorname{div} X = 0$ by the $H^1$ Green formula. On the mean-zero subspace of $H^1(\Omega)$, the Poincaré inequality makes the Dirichlet form coercive. The Lax–Milgram theorem therefore gives a unique mean-zero $p \in H^1(\Omega)$ satisfying

$$\int_\Omega \nabla p \cdot \nabla \varphi = \int_\Omega X \cdot \nabla \varphi = -\int_\Omega (\operatorname{div} X)\varphi.$$

The equality with the last integral again uses the zero normal trace of $X$. Compatibility extends the identity to every $\varphi \in H^1(\Omega)$. Since $\operatorname{div} X \in L^2(\Omega)$ and its integral is zero, $H^2$ regularity for the compatible homogeneous Neumann problem on a smooth domain gives $p \in H^2(\Omega)$, with $\Delta p = \operatorname{div} X$ and $\partial_\nu p = 0$. Put $W = X - \nabla p$. Then $W \in \mathcal{X}(\Omega)$ and $\operatorname{div} W = 0$. The weak construction also gives

$$\int_\Omega W \cdot \nabla \varphi = 0 \qquad (\varphi \in H^1(\Omega)). \tag{2.5}$$

Taking $\varphi = p$ proves

$$\int_\Omega W \cdot \nabla p = 0, \qquad \|X\|_{L^2}^2 = \|\nabla p\|_{L^2}^2 + \|W\|_{L^2}^2.$$

The divergence and antisymmetric derivatives also separate:

$$\mathcal{Q}(X) = \|\Delta p\|_{L^2}^2 + \mathcal{Q}(W). \tag{2.6}$$

The gradient part satisfies

$$\|\nabla p\|_{L^2}^2 = -\int_\Omega p \Delta p \leq \|p\|_{L^2}\|\Delta p\|_{L^2} \leq \mu^{-1/2}\|\nabla p\|_{L^2}\|\Delta p\|_{L^2}.$$

Consequently

$$\|\Delta p\|_{L^2}^2 \geq \mu \|\nabla p\|_{L^2}^2. \tag{2.7}$$

Suppose equality holds in (2.7). If $\nabla p = 0$, its zero mean gives $p = 0$. Otherwise the preceding chain has equal endpoints and forces equality in (2.1). Thus $p \in E_1(\Omega)$ in either case.

The key point for the remaining field is that every Cartesian component of $W$ has zero mean. Take $\varphi = x_i$ in (2.5) to obtain

$$\int_\Omega W_i = \int_\Omega W \cdot \nabla x_i = 0. \tag{2.8}$$

Apply the scalar Poincaré inequality to each $W_i$ and use convexity in (2.3). This yields

$$\mathcal{Q}(W) = \int_\Omega |DW|^2 + \int_{\partial\Omega} \mathrm{II}(W, W) \geq \mu \int_\Omega |W|^2. \tag{2.9}$$

Combining (2.6)– (2.9) proves (2.4).

Suppose equality holds. Both inequalities (2.7) and (2.9) must then be equalities. In the latter, all componentwise Poincaré inequalities are equalities and

$$\int_{\partial\Omega} \mathrm{II}(W, W) = 0.$$

Thus every $W_i$ belongs to $E_1(\Omega)$. The final assertion of Lemma 2.2 applies to the tangent field $W$ and gives $W = 0$. Equality in the gradient inequality then gives $X = \nabla p$ with $p \in E_1(\Omega)$.

Conversely, if $p \in E_1(\Omega)$, its gradient has tangent trace and zero antisymmetric derivative, and

$$\mathcal{Q}(\nabla p) = \int_\Omega (\Delta p)^2 = \mu^2 \int_\Omega p^2 = \mu \int_\Omega |\nabla p|^2.$$

This proves the equality characterization. □

**Lemma 2.4** (Injectivity of the mean gradient)**.** *The linear map*

$$J : E_1(\Omega) \longrightarrow \mathbb{R}^n, \qquad J(u) = \int_\Omega \nabla u$$

*is injective. More precisely,*

$$\frac{\mu}{|\Omega|}|J(u)|^2 \geq \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u) \qquad (u \in E_1(\Omega)). \tag{2.10}$$

*Proof.* Apply (2.3) to the tangent field $\nabla u$. The last calculation in Lemma 2.3 gives

$$\int_\Omega |D^2 u|^2 + \int_{\partial\Omega} \mathrm{II}(\nabla_\tau u, \nabla_\tau u) = \mu \int_\Omega |\nabla u|^2.$$

On the other hand, applying (2.1) to each $\partial_i u$ and adding gives

$$\int_\Omega |D^2 u|^2 \geq \mu \int_\Omega |\nabla u|^2 - \frac{\mu}{|\Omega|}|J(u)|^2.$$

The two formulas prove (2.10).

If $J(u) = 0$, each $\partial_i u$ has zero mean. The preceding two formulas and convexity force the curvature integral to vanish and every componentwise Poincaré inequality to be an equality. Thus $\partial_i u \in E_1(\Omega)$ by the scalar equality characterization, including its Neumann boundary condition. The final assertion of Lemma 2.2, now applied to the tangent field $\nabla u$, gives $\nabla u = 0$. The zero mean of $u$ then gives $u = 0$, proving injectivity. □

**Corollary 2.5** (Symmetry of the first eigenspace)**.** *One has* $\dim E_1(\Omega) \leq n$*. If an orthogonal map* $S$ *preserves* $\Omega$*, and* $(T_S u)(x) = u(S^{-1}x)$*, then*

$$J(T_S u) = S J(u). \tag{2.11}$$

*If* $\Omega = -\Omega$*, every first eigenfunction is centrally odd:*

$$u(-x) = -u(x). \tag{2.12}$$

*Proof.* The dimension bound follows from injectivity. Orthogonal changes of variables preserve the weak Neumann equation, so $T_S$ maps $E_1(\Omega)$ to itself. Since $\nabla(T_S u)(x) = S\nabla u(S^{-1}x)$ and $|\det S| = 1$, integration gives (2.11). For $S = -I$, this yields $J(T_{-I}u) = -J(u) = J(-u)$. Injectivity gives $T_{-I}u = -u$, which is (2.12). □

## 3. The first eigenspace and the axial variational proof

Assume (G1)–(G3) throughout this section, and let $d = n - 1$. On the nonpolar part of the curved boundary, an outward unit normal is

$$\nu = \frac{e_\rho - r'(z)e_z}{\sqrt{1 + r'(z)^2}}. \tag{3.1}$$

In particular $\nu_\rho > 0$, and $z\nu_z > 0$ when $z \neq 0$. The Neumann condition for a rotational function is therefore

$$U_\rho = r'(z)U_z. \tag{3.2}$$

### 3.1. **Odd ground states and spectral classification.**

**Lemma 3.1** (Coordinate-odd ground states)**.** *The lowest eigenvalue in each coordinate-odd subspace is simple in that subspace. Its eigenfunction can be chosen positive on the corresponding positive half-domain, is even in every other principal coordinate, and is smooth on* $\overline{\Omega}$*.*

*Proof.* Restriction to the positive half-domain identifies the odd variational problem with a mixed problem: Dirichlet data on the central flat section and Neumann data on the exterior boundary. More precisely, its form domain consists of the $H^1$ functions on that half-domain whose trace vanishes on the flat section; odd extension across that section is an $H^1$ function on $\Omega$ and multiplies both quadratic integrals by two; the matching zero traces prevent any jump across the reflection plane. This extension has zero mean, so (2.1) bounds its quotient below by $\mu_1 > 0$. The half-domain is bounded, convex and connected, hence Lipschitz. Compactness of its $H^1$ embedding in $L^2$ and weak closedness of the zero-trace condition give a normalized minimizer. Replacing a minimizer by its absolute value preserves its quotient; interior elliptic regularity and the strong minimum principle make this minimizer strictly positive in the connected interior.

Every lowest eigenfunction has one sign: otherwise its absolute value, also a smooth positive interior minimizer, would vanish at an interior point. Two linearly independent lowest eigenfunctions would have a nonzero linear combination orthogonal to a positive one, contradicting this sign property. This proves simplicity.

The odd extension is a weak Neumann eigenfunction on the full smooth domain. To check this, split any test function into its odd and even parts; the odd part restricts to an admissible mixed test, and the even part has zero pairing by parity. Full-domain Neumann regularity therefore gives smoothness up to the boundary, including the intersection of the reflection plane with the curved boundary. Reflections in all other coordinates preserve the positive normalized mixed ground state, so simplicity makes it even in those coordinates. □

Let $U$ be the positive-on-$z > 0$ axial ground state. Every orthogonal transformation of $y$ preserves its mixed problem and its positive $L^2$ normalization. Simplicity therefore gives $U = U(\rho, z)$; when $d = 1$, the transverse reflection supplies the same conclusion. Let $u_i$, $1 \le i \le d$, be the transverse ground states, positive on $y_i > 0$. Their eigenvalues are all $\lambda_\perp$.

**Proposition 3.2** (The axial and transverse alternatives)**.** *One has* $\mu_1 = \min\{\lambda_\parallel, \lambda_\perp\}$*, and*

$$E_1(\Omega) = \begin{cases} \operatorname{span}\{U\}, & \lambda_\parallel < \lambda_\perp, \\ \operatorname{span}\{u_1, \dots, u_d\}, & \lambda_\perp < \lambda_\parallel, \\ \operatorname{span}\{U, u_1, \dots, u_d\}, & \lambda_\parallel = \lambda_\perp. \end{cases} \tag{3.3}$$

*Whenever the transverse modes belong to* $E_1(\Omega)$*, they can be normalized so that*

$$(u_1(y,z), \dots, u_d(y,z)) = h(\rho^2, z)y, \qquad h(\rho^2, z) > 0 \quad (0 < \rho < r(z)), \tag{3.4}$$

*where* $h$ *is smooth at the axis and even in* $z$*.*

*Proof.* Coordinate reflections act on $E_1(\Omega)$ as commuting self-adjoint involutions. A joint parity component $f$ can have $\int_\Omega \partial_i f \neq 0$ only if it is odd in coordinate $i$ and even in all the others. All other parity patterns have zero mean gradient and hence vanish by Lemma 2.4. Every surviving component attains the lowest Rayleigh quotient in its odd subspace: its quotient is $\mu_1$, whereas every odd function has mean zero and quotient at least $\mu_1$. Lemma 3.1 therefore identifies it with its odd ground state. At least one such component is nonzero, while $\mu_1 \le \lambda_\parallel$ and $\mu_1 \le \lambda_\perp$ by the same mean-zero variational principle. This proves the minimum formula. Every ground state whose odd-sector eigenvalue equals $\mu_1$ is itself in $E_1(\Omega)$, so the parity decomposition proves exactly the direct sum (3.3).

Suppose the transverse modes are first modes. Parity gives $J(u_i) = c_i e_i$. Testing their full-domain weak eigenvalue equation with $y_i \in H^1(\Omega)$ gives

$$c_i = \int_\Omega \partial_i u_i = \lambda_\perp \int_\Omega y_i u_i > 0.$$

The last inequality follows from the chosen sign of $u_i$ on the positive half-domain and its oddness. Positive rescaling therefore normalizes the modes so that $J(u_i) = e_i$.

For $S \in O(d)$ acting on $y$ and fixing $z$, equivariance and injectivity of $J$ give $T_S u_i = \sum_j S_{ji} u_j$, since both sides have mean gradient $Se_i$. For $\Phi = (u_1, \dots, u_d)$ this is exactly $\Phi(Sy, z) = S\Phi(y, z)$. The subgroup fixing a nonzero $y$ then forces $\Phi(y, z)$ to be parallel to $y$, and rotations make its coefficient a function only of $\rho$ and $z$. When $d = 1$, the same conclusion follows from oddness in $y_1$.

To justify smoothness also at boundary points of the axis, first take a smooth extension $\widetilde{\Phi}$ to a neighborhood of $\overline{\Omega}$. By compactness we may restrict this extension to a sufficiently small neighborhood $\{x : \operatorname{dist}(x, \overline{\Omega}) < \varepsilon\}$, which is invariant under transverse orthogonal maps and $z \mapsto -z$. Replace the extension on that neighborhood by

$$\widehat{\Phi}(y, z) = \frac{1}{2} \int_{O(d)} S^{-1} \big( \widetilde{\Phi}(Sy, z) + \widetilde{\Phi}(Sy, -z) \big)\, dS,$$

where $dS$ is normalized Haar measure. The transformation law just proved and evenness in $z$ show that this average agrees with $\Phi$ on $\overline{\Omega}$; it is smooth, equivariant, and even in $z$. Set $g(t, z) = \widehat{\Phi}_1(te_1, z)$. It is smooth and odd in $t$, so

$$\frac{g(t, z)}{t} = \int_0^1 \partial_t g(\tau t, z)\, d\tau$$

extends smoothly as an even function of $t$. Its Taylor expansion through degree $2N$ has only even powers, and the remainder after $k$ derivatives in $s = t^2$ is $O(s^{N+1-k})$, by the chain rule and Taylor's differentiated remainder estimates. The same estimates hold after any fixed number of $z$ derivatives. Since $N$ is arbitrary, substitution $t = \sqrt{s}$ gives smooth one-sided derivatives of every order at $s = 0$, jointly with $z$. This gives the required smooth $h(\rho^2, z)$, including at the poles; equivariance reconstructs the full vector. The same even-function argument applies to smooth rotationally invariant scalar functions. Finally, positivity of $u_1(\rho e_1, z)$ for $0 < \rho < r(z)$ gives the stated interior positivity of $h$, and evenness in $z$ is preserved throughout. □

### 3.2. **Changing the signs of a meridional gradient.**

**Proposition 3.3** (Axial monotonicity)**.** *If $\lambda_\parallel = \mu_1(\Omega)$, then its positive axial ground state satisfies*

$$U_z > 0 \quad \text{in } \Omega, \qquad U_\rho < 0 \quad \text{for } z > 0,\ \rho > 0. \tag{3.5}$$

*Its unique maximum and minimum on $\overline{\Omega}$ are $P_+$ and $P_-$. Moreover, $U_z > 0$ on $\partial\Omega \setminus \{P_+, P_-\}$. The assertion holds for the axial mode even if the transverse eigenvalue is equal to it.*

*Proof.* At $\rho > 0$ put $p = U_z$ and $q = U_\rho$, and define the vector field

$$X = |p| e_z - \operatorname{sgn}(z) |q| e_\rho. \tag{3.6}$$

We first justify its regularity across the sets where these coordinates or signs are singular. By the smooth even-function argument in Proposition 3.2, applied to a smooth extension averaged over transverse orthogonal maps and antisymmetrized in $z$, write $U(\rho, z) = H(\rho^2, z)$ with $H$ smooth and odd in $z$. Smooth division of $2H_s$ by $z$ gives

$$q(\rho, z) = \rho z K(\rho^2, z)$$

with $K$ smooth, including at the axis and equatorial plane. The transverse part of (3.6) is therefore exactly $-z|K(|y|^2, z)|y$. Together with $X_z = |U_z|$, this defines a locally Lipschitz field near $\overline{\Omega}$,

and in particular an $H^1$ field. A smooth symmetric extension of $U$ justifies the same statement at the poles and the boundary.

The Neumann relation (3.2), and the sign of $r'$ on each half, show on the nonpolar boundary that

$$X = \operatorname{sgn}(p)\nabla U$$

where $p \neq 0$; both fields are zero there if $p = 0$. The same identity for the quadratic boundary quantities holds at the equator and poles by continuity. Thus $X \cdot \nu = 0$ and $\mathrm{II}(X, X) = \mathrm{II}(\nabla U, \nabla U)$ on the boundary.

For a meridional field $A(\rho, z)e_\rho + B(\rho, z)e_z$, direct differentiation in an orthonormal polar frame gives

$$|DX|^2 = A_\rho^2 + A_z^2 + B_\rho^2 + B_z^2 + (n-2)\frac{A^2}{\rho^2}.$$

Applied to (3.6), the chain rule for absolute values shows, almost everywhere, that $|DX|^2 = |D^2U|^2$ and $|X|^2 = |\nabla U|^2$. The $H^1$ justification above ensures that no distribution supported on the axis or equatorial plane is omitted. The Gaffney identity now gives

$$\mathcal{Q}(X) = \mathcal{Q}(\nabla U) = \mu_1 \int_\Omega |\nabla U|^2 = \mu_1 \int_\Omega |X|^2.$$

By Lemma 2.3, $X = \nabla V$ for some $V \in E_1(\Omega)$. The field $X$ is equivariant under every transverse orthogonal transformation $S$. Consequently $\nabla(V \circ S) = S^T X(Sx) = X(x) = \nabla V$. The difference $V \circ S - V$ is constant on the connected domain, and its zero mean makes that constant zero. Thus $V$ is invariant under all these transformations, including the reflection when $d = 1$. Proposition 3.2 makes that invariant subspace exactly $\operatorname{span}\{U\}$. Hence $X = c\nabla U$, and equality of norms gives $c = 1$ or $c = -1$. The latter would make $U_z \leq 0$ everywhere, contrary to $U(0,0) = 0$ and $U(0, z) > 0$ for $0 < z < L$. Therefore $c = 1$, proving the nonstrict signs in (3.5).

The function $p = U_z \geq 0$ is not identically zero, since $U(0,0) = 0$ and $U(0, z) > 0$ for $0 < z < L$. It satisfies $-\Delta p = \mu_1 p \geq 0$. The strong minimum principle yields $p > 0$ in the entire domain. On the connected meridian region $0 < \rho < r(z)$, $0 < z < L$, the function $w = -U_\rho \geq 0$ satisfies

$$w_{\rho\rho} + \frac{n-2}{\rho} w_\rho + w_{zz} + \left(\mu_1 - \frac{n-2}{\rho^2}\right) w = 0.$$

On any sufficiently small neighborhood away from $\rho = 0$, its coefficients are bounded. Choose a constant $C \leq \min\{0, \inf(\mu_1 - (n-2)/\rho^2)\}$ on that neighborhood. Then

$$w_{\rho\rho} + \frac{n-2}{\rho} w_\rho + w_{zz} + Cw \leq 0.$$

The strong minimum principle, applied locally and then using connectedness, makes $w$ either positive throughout the meridian region or identically zero. The latter would make $U$ independent of $\rho$ on the upper half; the Neumann condition and $r'(z) < 0$ would then give $U_z(z) = 0$ for every $0 < z < L$, contradicting the already proved positivity of $p$. Thus $w > 0$. The same local supersolution inequality also permits Hopf's boundary point lemma at every nonpolar upper boundary point.

We also prove the asserted boundary positivity of $p = U_z$. At a nonpolar upper boundary point, $\nu_\rho, \nu_z > 0$. If $p = 0$ there, the Neumann relation gives $q = 0$, hence $\nabla U = 0$. For every tangent vector $T$, differentiating the Neumann condition at this critical point gives $0 = D^2U(T, \nu) + \nabla U \cdot D_T\nu = D^2U(T, \nu)$. Symmetry of the Hessian therefore gives

$$D^2U\,\nu = A\nu, \qquad A = U_{\nu\nu}.$$

Hopf's boundary point lemma applied to $p > 0$ yields $\partial_\nu p = A\nu_z < 0$, and hence $A < 0$. Applied in the smooth meridian region to $w = -U_\rho > 0$, it yields $\partial_\nu w = -A\nu_\rho < 0$, and hence $A > 0$, a contradiction. At an equatorial boundary point, $q = 0$ by oddness of $U$ and $\nu_z = 0$. A hypothetical

zero of $p$ would again imply $\nabla U = 0$ and then $\partial_\nu p = A\nu_z = 0$, contradicting Hopf's lemma for $p$. Reflection gives the lower half. At the two poles $p = 0$ by the Neumann condition, as expected.

To locate the extrema, first take a point in the closed upper half-domain with $\rho > 0$. Strict radial decrease, applied first on an interior subsegment and then by continuity to a possible boundary endpoint, gives $U(\rho, z) < U(0, z)$ when $z > 0$. Strict axial increase similarly gives $U(0, z) < U(0, L)$ for $0 \le z < L$. The equatorial values are zero and the values on the negative half are nonpositive. Hence the unique maximum is $P_+$; oddness proves the unique minimum at $P_-$. □

### 3.3. **A transverse slice barrier.**

**Theorem 3.4** (Axial spectral selection)**.** *If*

$$\lambda_\parallel < \frac{\beta_{n-1}^2}{R^2}, \tag{3.7}$$

*then $\lambda_\parallel < \lambda_\perp$, and the full first eigenspace has all the axial conclusions of Theorem 1.1.*

*Proof.* For any $y_1$-odd function $f \in H^1(\Omega)$, almost every transverse slice has zero mean on $B^{n-1}_{r(z)}$. Slice Poincaré and Fubini give

$$\int_\Omega |\nabla f|^2 \ge \int_{-L}^{L} \frac{\beta_{n-1}^2}{r(z)^2} \int_{B^{n-1}_{r(z)}} f(y,z)^2 \, \mathrm{d}y \, \mathrm{d}z \ge \frac{\beta_{n-1}^2}{R^2} \int_\Omega f^2.$$

For precision, first apply the slice inequality to smooth approximants on the truncated domains $\Omega \cap \{|z| < L - \varepsilon\}$. Antisymmetrization preserves the coordinate-odd condition, and the weight $r(z)^{-2}$ is bounded on each such domain. Thus $H^1$ convergence passes the inequality to $f$ there. Letting $\varepsilon \downarrow 0$ by monotone convergence justifies the display on the whole domain and proves finiteness of its weighted integral, despite the singular weight at the poles. In fact the bound for the lowest transverse eigenvalue is strict:

$$\lambda_\perp > \frac{\beta_{n-1}^2}{R^2}.$$

To see this, apply the displayed inequality to its nonzero ground state, whose existence was proved in Lemma 3.1. The strict decrease of $r$ on $(0, L)$ and its evenness give $r(z) < R$ for every $z \neq 0$. A nonzero $L^2$ function has positive slice mass on a set of $z$ of positive measure, so the second inequality in the display is strict for that ground state. Attainment is used here: strict inequalities for individual trial functions alone would not imply a strict bound on their infimum. Combine this with (3.7), Proposition 3.2, and Proposition 3.3. □

## 4. The transverse first modes and flat bodies

Use the notation of Proposition 3.2, with $x = (y, z) \in \mathbb{R}^d \times \mathbb{R}$, $d = n - 1$, $\rho = |y|$, and $R = r(0)$. The eigenvalues $\lambda_\parallel$ and $\lambda_\perp$ are the lowest eigenvalues in the axial-odd and a transverse-coordinate-odd subspace, respectively. In particular, $\mu_1(\Omega) = \min\{\lambda_\parallel, \lambda_\perp\}$. Assume (G1)–(G3) throughout this section.

The upper meridional boundary can also be written as $z = f(\rho)$. Here $f$ is the inverse of $r$ restricted to $(0, L)$, extended by $f(0) = L$ and $f(R) = 0$. The inverse function theorem gives, on $(0, R)$,

$$f'(\rho) = \frac{1}{r'(f(\rho))} < 0, \qquad f''(\rho) = -\frac{r''(f(\rho))}{r'(f(\rho))^3} \le 0.$$

Thus $f$ is decreasing and concave in every dimension under consideration.

**Proposition 4.1** (Strict monotonicity of a transverse first mode)**.** *Suppose $\lambda_\perp = \mu_1(\Omega) = \mu$. For a unit vector $e \in \mathbb{R}^d$, let $u_e$ be the corresponding transverse mode, chosen positive when $y \cdot e > 0$. Then*

$$\partial_e u_e > 0 \quad \text{in } \Omega \quad \text{and on } \partial\Omega \setminus \{(Re, 0), (-Re, 0)\}. \tag{4.1}$$

*Its unique maximum and minimum are attained at $(Re, 0)$ and $(-Re, 0)$, respectively. Writing*

$$u_e(y, z) = v(\rho, z)\frac{y \cdot e}{\rho}, \qquad v(\rho, z) > 0 \quad (0 < \rho < r(z)),$$

*one has*

$$v_\rho > 0 \quad \text{in the meridional interior with } \rho > 0, \qquad v_z < 0 \quad \text{there when } z > 0. \tag{4.2}$$

*These assertions apply to each transverse mode even if $\lambda_\perp = \lambda_\parallel$; they do not assert a conclusion for a mixture of the two branches at such a tie.*

*Proof.* A transverse orthogonal change of coordinates reduces the proof to $e = e_1$ in every dimension, including $d = 1$. First assume $d \geq 2$, and put

$$Y(\theta) = \theta_1, \qquad e_\rho = \theta = \frac{y}{\rho}, \qquad a = v_\rho, \quad b = \frac{v}{\rho}, \quad c = v_z.$$

Proposition 3.2 gives

$$v(\rho, z) = \rho h(\rho^2, z), \qquad h(\rho^2, -z) = h(\rho^2, z). \tag{4.3}$$

The same proposition supplies smoothness of $h$ up to the axis and the boundary, including the poles, and its evenness in $z$. This factorization is obtained from the smooth equivariant extension to a neighborhood of the closed domain constructed in that proof; it is consequently legitimate to differentiate it at a boundary point of the axis. In particular, $b = h > 0$ in the interior when $\rho > 0$, and $b \geq 0$ on the closure by continuity. The gradient is

$$\nabla u = aYe_\rho + b\nabla_S Y + cYe_z, \qquad \nabla_S Y = e_1 - Ye_\rho. \tag{4.4}$$

Define a vector field by

$$X = |a|Ye_\rho + b\nabla_S Y - \operatorname{sgn}(z)|c|Ye_z. \tag{4.5}$$

We first justify its regularity. Its transverse component is

$$X_y = be_1 + (|a| - b)Ye_\rho.$$

Equation (4.3) gives $a - b = 2\rho^2 h_s(\rho^2, z)$, where $s$ is the first variable of $h$. Since $b \geq 0$, the reverse triangle inequality yields

$$\big||a| - b\big| \leq |a - b| = O(\rho^2).$$

Thus the apparently singular angular term has a continuous extension through the axis. Its angular derivatives divided by $\rho$ are $O(\rho)$. Moreover $a_\rho, b_\rho = O(\rho)$, so the Lipschitz chain rule makes its radial derivative $O(\rho)$; its axial derivative is bounded. These estimates are uniform locally up to the boundary. There is no extra weak derivative supported on the axis: inserting a radial cutoff at distance $\varepsilon$ from it in the distributional derivative calculation gives an error $O(\varepsilon^{d-1})$, which tends to zero for every $d \geq 2$. The axial component has the nonsingular expression

$$X_z = -\operatorname{sgn}(z)|h_z(\rho^2, z)|y_1.$$

Smooth division by $z$ gives $h_z(s, z) = zk(s, z)$ with $k$ smooth. Thus $X_z = -z|k(\rho^2, z)|y_1$ is locally Lipschitz across $z = 0$ and the axis. A finite cover of $\overline{\Omega}$, using the smooth extensions already described at the poles, now gives $X \in H^1(\Omega; \mathbb{R}^n)$.

The three summands in (4.4) are orthogonal pointwise, so

$$|X|^2 = |\nabla u|^2. \tag{4.6}$$

For completeness, with $\nabla_m = (\partial_\rho, \partial_z)$, the Cartesian derivative energy of a field of the form on the right of (4.4) is

$$|D(\nabla u)|^2 = Y^2(|\nabla_m a|^2 + |\nabla_m c|^2) + (1-Y^2)|\nabla_m b|^2 + \frac{(a-b)^2(1+(d-2)Y^2) + c^2(1-Y^2)}{\rho^2}. \tag{4.7}$$

To verify the angular part, write the transverse field as $be_1 + (a-b)Ye_\rho$ and differentiate in an orthonormal tangent frame on $S^{d-1}$. For a unit tangent vector $\tau$,

$$D_\tau(Ye_\rho) = (e_1 \cdot \tau)e_\rho + Y\tau.$$

The two summands are orthogonal. Summing their squared norms over the frame gives $(1-Y^2)+(d-1)Y^2 = 1+(d-2)Y^2$. The axial field contributes $c^2|\nabla_S Y|^2 = c^2(1-Y^2)$. The meridional part follows directly from the orthogonality of $Ye_\rho$, $\nabla_S Y$, and $Ye_z$. Replacing $a, c$ as in (4.5) preserves the squared meridional weak derivatives almost everywhere and preserves $c^2$. Since $b > 0$, it gives

$$|D(\nabla u)|^2 - |DX|^2 = \frac{4b(-a)_+(1+(d-2)Y^2)}{\rho^2} \geq 0 \quad \text{almost everywhere.} \tag{4.8}$$

On the curved boundary the normal is meridional. Write its components as $(\nu_\rho, \nu_z)$; away from the poles, $\nu_\rho > 0$, and $\nu_z$ has the sign of $z$. Applying the Neumann condition at the representative point $(\rho e_1, z)$, where $Y = 1$, gives the meridional identity

$$\nu_\rho a + \nu_z c = 0. \tag{4.9}$$

For $z \neq 0$, the replacement of $(a, c)$ in (4.5) therefore multiplies both entries by the same sign, or leaves both zero. At $z = 0$ both entries vanish: $a = 0$ by (4.9) and $c = 0$ by reflection. The azimuthal component $b\nabla_S Y$ is unchanged. Hence $X$ is tangent to the boundary. On a surface of revolution the meridional tangent line and the azimuthal tangent space are orthogonal principal spaces of the second fundamental form: differentiating (3.1) in a meridional tangent direction $T$ keeps $D_T\nu$ in the meridional plane, whereas an azimuthal unit vector $E$ satisfies $D_E\nu = (\nu_\rho/\rho)E$. Thus the second fundamental form has no mixed term between these spaces. The replacement changes only the sign of the meridional tangent component, and consequently

$$\mathrm{II}(X, X) = \mathrm{II}(\nabla u, \nabla u) \quad \text{almost everywhere on } \partial\Omega. \tag{4.10}$$

The two poles have surface measure zero, so no coordinate assertion at those points is required for this equality.

The Gaffney identity and (4.8)– (4.10) imply

$$\mathcal{Q}(X) \leq \mathcal{Q}(\nabla u) = \mu\|\nabla u\|_{L^2}^2 = \mu\|X\|_{L^2}^2.$$

Lemma 2.3 forces equality and gives $X = \nabla p$ for a first eigenfunction $p$. The nonnegative right-hand side of (4.8) must therefore have zero integral. Since $d \geq 2$, its angular factor is at least one, and $b > 0$ forces $a \geq 0$ almost everywhere. Continuity away from the axis upgrades this to every meridional point. Thus the transverse components of $X$ and $\nabla u$ agree. Each transverse slice of $\Omega$ is a connected ball, so $p - u$ depends only on $z$. Its axial derivative is, for $\rho > 0$,

$$(-\operatorname{sgn}(z)|c| - c)Y.$$

A function independent of $y$ cannot have this nonconstant angular dependence: evaluate at $\theta = e_1$ and $\theta = -e_1$ on each slice. The displayed coefficient must vanish. This proves $c \leq 0$ for $z > 0$, without any assumption about simplicity of the full first eigenspace.

Now

$$w := \partial_{y_1} u = aY^2 + b(1-Y^2) \geq 0.$$

Continuity gives the same nonnegativity on the axis. The function is nonzero: since $d \geq 2$, there are interior points with $Y = 0$, where $w = b > 0$. It satisfies $-\Delta w = \mu w \geq 0$ in $\Omega$. The strong minimum principle therefore gives $w > 0$ throughout the interior. Evaluating at $\theta = e_1$ proves $a > 0$ for $\rho > 0$. Substitution of $u = vY$ in its eigenvalue equation, using $-\Delta_S Y = (d-1)Y$, gives the amplitude equation

$$v_{zz} + v_{\rho\rho} + \frac{d-1}{\rho} v_\rho + \left(\mu - \frac{d-1}{\rho^2}\right) v = 0.$$

Its coefficients are independent of $z$, so $-c = -v_z \geq 0$ solves the same equation in the connected upper meridional region $\{(\rho, z) : 0 < z < L,\ 0 < \rho < r(z)\}$. Away from $\rho = 0$, its coefficients are locally bounded. More explicitly, on any relatively compact neighborhood write the equation as $\mathcal{L}(-c) = 0$, with

$$\mathcal{L} = \partial_{\rho\rho} + \partial_{zz} + \frac{d-1}{\rho}\partial_\rho + \mu - \frac{d-1}{\rho^2}.$$

Replacing its zeroth-order coefficient by $-M$, where $M \geq 0$ is large enough that $-M \leq \mu - (d-1)/\rho^2$ on that neighborhood, makes the nonnegative function $-c$ a supersolution for an operator with nonpositive zeroth-order coefficient. The strong minimum principle therefore applies. Thus $-c$ is either positive throughout that region or identically zero. In the latter case write $v(\rho, z) = V(\rho)$ there. Each $0 < \rho < R$ occurs on the upper cap, where (4.9) gives $V'(\rho) = 0$, contradicting the already proved $a > 0$. Hence $c < 0$ in the upper meridional interior. At every curved boundary point with $\rho, z > 0$, the meridional boundary is smooth and has an interior ball, and $\rho$ stays bounded away from zero on a sufficiently small neighborhood. The same local supersolution argument therefore justifies the boundary point lemma for $-c$ below.

We give the boundary argument to rule out additional extrema. First, $v > 0$ on the meridional boundary wherever $\rho > 0$. Indeed, at the representative point $(\rho e_1, z)$ the eigenfunction is positive on the adjacent interior of the half-domain $y_1 > 0$; this point is away from its flat Dirichlet boundary. A zero boundary value there would contradict its Neumann condition by Hopf's boundary point lemma. Thus $b > 0$ at all these points.

Consider a point $(\rho e_1, z)$ on the boundary with $\rho, z > 0$. If $w = 0$ there, then $a = 0$ and (4.9) gives $c = 0$, so $\nabla u = 0$. The differentiated Neumann identity at a boundary critical point, established in the proof of Proposition 3.3, gives

$$D^2 u\, \nu = A\nu, \qquad A = u_{\nu\nu}.$$

Hopf's lemma for the positive function $w$ gives $\partial_\nu w = A\nu_\rho < 0$, hence $A < 0$. Apply Hopf's lemma also to the positive meridional function $-c$. Its outward meridional normal is $(\nu_\rho, \nu_z)$, and at the point in question

$$\partial_\nu(-c) = -A\nu_z < 0,$$

which gives $A > 0$, a contradiction. Reflections in $z$ and $y_1$ give the same conclusion at the other non-equatorial points with $\theta = \pm e_1$.

At a boundary point with $\rho > 0$ and $\theta \neq \pm e_1$, the expression for $w$ gives $w \geq b(1 - Y^2) > 0$. At an axial pole, (4.3) shows that all components of $\nabla u$ except possibly $w$ vanish. If $w = 0$ there, the preceding differentiated Neumann identity again applies; since $\nu \cdot e_1 = 0$, it would give $\partial_\nu w = 0$, contradicting Hopf's lemma. We have proved (4.1). At $(\pm Re_1, 0)$ the direction $e_1$ is normal and $w = 0$ by the Neumann condition.

An interior extremum is impossible because $w > 0$. At any boundary extremum, tangential stationarity and the Neumann condition imply $\nabla u = 0$. The boundary positivity just proved leaves only $(\pm Re_1, 0)$. Their values have opposite signs, with the positive value at $(Re_1, 0)$, proving the asserted uniqueness.

When $n = 2$, exchange the two coordinates. The domain is again a smooth symmetric convex body of revolution, now with axis $y_1$. Its new profile is the even extension of the inverse function

$f$ defined at the start of this section, which is decreasing and concave on $(0, R)$. At the original upper pole the normal is $e_z$, so the smooth boundary writes $z$ as a smooth even function of $y_1$ near zero. Thus this even extension is smooth at its equator, and the coordinate exchange preserves smoothness of the boundary. Together with the endpoint values of $f$, these facts show that the exchanged profile satisfies (G1)–(G3) as well. The mode under consideration is its axial-odd first mode and is even in its transverse coordinate. Proposition 3.3 therefore gives $\partial_{y_1} u > 0$ in the interior and on the boundary except at $(\pm R, 0)$, with these two points its unique extrema. For $y_1 > 0$ its axial derivative in the exchanged coordinates is $v_\rho$, while its radial transverse derivative is $v_z$ when $z > 0$. The two strict signs in (4.2) follow as well. □

4.1. **Explicit conditions selecting the transverse branch.** Define the transverse second moment, independent of the choice of unit $e \in \mathbb{R}^d$, by

$$M_\perp = \frac{1}{|\Omega|} \int_\Omega (y \cdot e)^2 \, dx.$$

**Proposition 4.2** (Transverse moment criterion)**.** *If*

$$M_\perp \geq \frac{4L^2}{\pi^2}, \tag{4.11}$$

*then $\lambda_\perp < \lambda_\parallel$. Every nonzero first eigenfunction is a transverse mode for a suitable direction and therefore satisfies Proposition 4.1. In particular, the first eigenvalue has multiplicity $n - 1$.*

*Proof.* The coordinate trial function $y \cdot e$ is odd in that coordinate and gives

$$\lambda_\perp < \frac{1}{M_\perp}.$$

The inequality is strict: equality would make this nonzero linear function an odd-sector ground state and hence, by Lemma 3.1, a full-domain Neumann eigenfunction with positive eigenvalue. Its zero Laplacian makes this impossible.

For almost every fixed $y$ with $|y| < R$, an $H^1$ function $w$ odd in $z$ has a continuous odd representative on its vertical interval, so its trace at $z = 0$ is zero. Applying the interval Poincaré inequality on $(-f(|y|), f(|y|))$ and using oddness gives

$$\int_0^{f(|y|)} |\partial_z w|^2 \, dz \geq \frac{\pi^2}{4f(|y|)^2} \int_0^{f(|y|)} |w|^2 \, dz \geq \frac{\pi^2}{4L^2} \int_0^{f(|y|)} |w|^2 \, dz.$$

No condition at $z = f(|y|)$ is imposed in this inequality. For the Sobolev passage at the degenerating endpoints, use smooth odd approximants on $|y| < R - \varepsilon$, where $f(|y|)^{-2}$ is bounded, and then monotone convergence as in the proof of Theorem 3.4. Integration and odd reflection give $\lambda_\parallel \geq \pi^2/(4L^2)$. Consequently (4.11) implies $\lambda_\perp < \lambda_\parallel$. The first-eigenspace description and multiplicity follow from Proposition 3.2. □

**Corollary 4.3** (An elementary flat-body criterion)**.** *The sufficient condition in Proposition 4.2 holds whenever*

$$\frac{L}{R} \leq \frac{\pi}{2} \sqrt{\frac{n}{(n+1)(n+2)}}. \tag{4.12}$$

*Proof.* The concavity and endpoint values of $f$ established at the start of the section give the tangent-line inequality

$$f(\rho) + (R - \rho) f'(\rho) \geq 0,$$

so $q(\rho) = f(\rho)/(R-\rho)$ is nondecreasing on $(0, R)$. Relative to the probability measure proportional to $\rho^{d-1}(R - \rho) \, d\rho$, the functions $q(\rho)$ and $\rho^2$ therefore have nonnegative covariance. This follows

directly by integrating $(q(s)-q(t))(s^2-t^2)\geq 0$ against two copies of that measure. All the required moments are finite, because multiplication by $q$ replaces its density by $\rho^{d-1}f(\rho)$. The angular average of $(\theta\cdot e)^2$ is $1/d$, also for $d=1$. Polar integration therefore gives

$$\begin{aligned} M_\perp &= \frac{1}{d}\,\frac{\int_0^R \rho^{d+1}f(\rho)\,d\rho}{\int_0^R \rho^{d-1}f(\rho)\,d\rho} \\ &\geq \frac{1}{d}\,\frac{\int_0^R \rho^{d+1}(R-\rho)\,d\rho}{\int_0^R \rho^{d-1}(R-\rho)\,d\rho} = \frac{d+1}{(d+2)(d+3)}R^2 = \frac{n}{(n+1)(n+2)}R^2. \end{aligned}$$

Condition (4.12) therefore implies $M_\perp \geq 4L^2/\pi^2$, as required. □

## 5. Geometric conditions that select the axial eigenfunction

Assume (G1)–(G3) and retain the geometric and spectral notation of the preceding sections. In particular, $\lambda_{\mathrm{odd}}=\lambda_{\parallel}$ and $\beta_m^2=\mu_1(B^m)$ for the unit $m$-ball, with $\beta_1=\pi/2$. The purpose of this section is to give explicit sufficient conditions for

$$\lambda_{\mathrm{odd}} < \frac{\beta_{n-1}^2}{R^2}. \tag{5.1}$$

The spectral selection theorem, Theorem 3.4, then identifies this axial eigenvalue with the simple first positive Neumann eigenvalue of $\Omega$.

### 5.1. **A lower bound for the transverse ball eigenvalue.**

**Lemma 5.1.** *For every integer $m\geq 1$,*

$$\beta_m^2 > m+1. \tag{5.2}$$

*Proof.* Project the unit sphere $S^{m+1}\subset\mathbb{R}^{m+2}$ onto its first $m$ coordinates. Away from a set of surface measure zero, a parametrization is

$$(x,\sqrt{1-|x|^2}\cos\vartheta,\sqrt{1-|x|^2}\sin\vartheta),\qquad x\in B^m,\quad 0\leq\vartheta<2\pi.$$

The metric blocks in these coordinates are

$$g_{xx}=I+\frac{x\otimes x}{1-|x|^2},\qquad g_{x\vartheta}=0,\qquad g_{\vartheta\vartheta}=1-|x|^2.$$

The $x$ block has determinant $(1-|x|^2)^{-1}$ and inverse $I-x\otimes x$. Thus the full metric has determinant one, and the induced measure is $\mathrm{d}x\,\mathrm{d}\vartheta$. For smooth $f$ on $\overline{B^m}$, its lift $\widetilde{f}=f(x)$ is smooth on the entire sphere: extend $f$ smoothly past the closed ball and compose with the ambient coordinate projection. In particular, the collapsed circle fibers create no singularity in this function. The metric calculation gives

$$\int_{S^{m+1}}\widetilde{f}^2=2\pi\int_{B^m}f^2,\qquad \int_{S^{m+1}}|\nabla_S\widetilde{f}|^2=2\pi\int_{B^m}(|\nabla f|^2-(x\cdot\nabla f)^2).$$

The measure identity also shows that a mean-zero $f$ has a mean-zero lift. The spherical harmonic eigenvalues on $S^{m+1}$ are $\ell(\ell+m)$, $\ell=0,1,\ldots$, as follows from the polar-coordinate formula for the Euclidean Laplacian on homogeneous harmonic polynomials. Their restrictions form a complete orthogonal decomposition of $L^2(S^{m+1})$: harmonic decomposition expresses every polynomial restriction as a sum of such harmonics, and polynomial restrictions are dense. If $\widetilde{f}_\ell$ is the degree-$\ell$ component of a smooth mean-zero lift, integration by parts and Parseval's identity therefore give

$$\int_{S^{m+1}}|\nabla_S\widetilde{f}|^2=\sum_{\ell\geq 1}\ell(\ell+m)\|\widetilde{f}_\ell\|_2^2\geq(m+1)\int_{S^{m+1}}\widetilde{f}^2.$$

Combining this gap with the two lift identities gives

$$\int_{B^m} |\nabla f|^2 \geq (m+1) \int_{B^m} f^2 + \int_{B^m} (x \cdot \nabla f)^2.$$

For a mean-zero $f \in H^1(B^m)$, choose smooth approximants on the closed ball and subtract their means. Such approximants are obtained by the usual Sobolev extension from the smooth ball followed by mollification. They converge in $H^1$, and all three ball integrals in the last inequality converge, since $|x| \leq 1$. This extends the inequality to every such $f$. Apply it to a nonzero first Neumann eigenfunction on the ball. Equality $\beta_m^2 = m+1$ would force $x \cdot \nabla f = 0$. Smoothness then makes $f$ constant along every radial segment and continuity at the origin makes all these constants equal. This contradicts its nonzero, mean-zero normalization. The argument includes $m = 1$. □

5.2. **The axial second moment.**

**Proposition 5.2** (Moment criterion)**.** *If*

$$\frac{1}{|\Omega|} \int_\Omega z^2 \, \mathrm{d}x \geq \frac{R^2}{\beta_{n-1}^2}, \tag{5.3}$$

*then* (5.1) *holds, including equality in* (5.3)*.*

*Proof.* The odd trial function $v(y, z) = z$ gives the strict bound

$$\lambda_{\text{odd}} < \frac{|\Omega|}{\int_\Omega z^2 \, \mathrm{d}x}. \tag{5.4}$$

Strictness is the coordinate-trial argument of Proposition 4.2, applied to the axial coordinate $z$ instead of a transverse coordinate. Combining (5.4) and (5.3) proves the assertion. □

5.3. **Comparison with the double cone.** Let

$$\alpha_n^2 = \lambda_1^D(B^n), \qquad \alpha_n = j_{n/2-1,1}, \tag{5.5}$$

where $\lambda_1^D$ is the first Dirichlet eigenvalue and $j_{\nu,1}$ is the first positive zero of the Bessel function $J_\nu$. Only the variational definition and its radial ordinary differential equation are needed below.

**Lemma 5.3** (Double-cone trial)**.** *For the domain under consideration,*

$$\lambda_{\text{odd}} < \frac{\alpha_n^2}{L^2}. \tag{5.6}$$

*Proof.* The compactness, absolute-value and simplicity arguments of Lemma 3.1, now applied to $H_0^1(B^n)$, give a simple positive Dirichlet ground state. Rotational invariance and simplicity make it radial. Dirichlet regularity makes its radial profile $F$ smooth on $[0, 1]$; normalize it by $F(0) = 1$. It satisfies

$$F''(s) + \frac{n-1}{s} F'(s) + \alpha_n^2 F(s) = 0, \qquad F'(0) = 0, \qquad F(1) = 0.$$

For completeness, putting $\nu = n/2 - 1$ and $F(s) = s^{-\nu} G(\alpha_n s)$ reduces this equation to Bessel's equation of order $\nu$. The solution regular at zero is a positive multiple of $s^{-\nu} J_\nu(\alpha_n s)$, with its value at zero understood by continuity. Positivity of $F$ before its first boundary zero therefore gives $\alpha_n = j_{\nu,1}$, as stated in (5.5). In particular, $F > 0$ on $[0, 1)$. Integrating $(s^{n-1} F')' = -\alpha_n^2 s^{n-1} F$ from zero gives

$$F'(s) = -\alpha_n^2 s^{1-n} \int_0^s t^{n-1} F(t) \, \mathrm{d}t < 0 \qquad (0 < s \leq 1).$$

Set

$$V(z) = F(1 - z/L), \qquad 0 \leq z \leq L.$$

Then $V(0) = 0$, $V'(L) = 0$, and $V, V' > 0$ on $(0, L)$. Its equation is

$$-V''(z) + \frac{n-1}{L-z}V'(z) = \frac{\alpha_n^2}{L^2}V(z). \tag{5.7}$$

Since a differentiable concave function lies below each tangent line, $r(L) = 0$ implies

$$r(z) + (L-z)r'(z) \geq 0, \qquad \frac{r'(z)}{r(z)} \geq -\frac{1}{L-z} \quad (0 < z < L).$$

With $w(z) = r(z)^{n-1}$, equation (5.7) therefore yields

$$-\frac{(wV')'}{w} = -V'' - (n-1)\frac{r'}{r}V' \leq \frac{\alpha_n^2}{L^2}V. \tag{5.8}$$

This inequality is strict on an interval immediately to the right of 0: smoothness and reflection symmetry at the equator give $r'(0) = 0$, whereas $-1/(L-z)$ tends to $-1/L$.

Multiply (5.8) by $wV$ and integrate. The boundary term $wVV'$ vanishes at both endpoints: $V(0) = 0$, and $w(L) = 0$ while $V, V'$ are bounded. Integration can first be performed on $[0, L-\varepsilon]$ and then passed to the limit; this justifies it also when the profile graph has an unbounded derivative at the pole. Indeed, $w$ is decreasing, is locally absolutely continuous, and $\int_0^L |w'| \, \mathrm{d}z = w(0)$; hence the term $w'VV'$ is integrable. The other terms are integrable because $V, V', V''$ are bounded. The weighted deficit is positive on a fixed closed subinterval of the strictness region near zero. This interval is retained for all sufficiently small $\varepsilon$, so strictness survives the limit. The result is

$$\int_0^L w(V')^2 \, \mathrm{d}z < \frac{\alpha_n^2}{L^2}\int_0^L wV^2 \, \mathrm{d}z.$$

Oddly extend $V$ to $[-L, L]$ and regard it as a function on $\Omega$. Since $V(0) = 0$ and $V'$ is bounded, this extension is Lipschitz and belongs to $H^1(\Omega)$. Fubini's theorem cancels the common factor $2\omega_{n-1}$ in its two Rayleigh integrals, giving exactly the preceding weighted quotient and proving the claim. □

**Proposition 5.4** (Explicit aspect-ratio criteria)**.** *Either of the following conditions implies* (5.1)*:*

$$\frac{L}{R} \geq \frac{\alpha_n}{\beta_{n-1}}, \tag{5.9}$$

$$\frac{L}{R} \geq \sqrt{\frac{n+4}{2}}. \tag{5.10}$$

*Proof.* The exact condition (5.9), combined with Lemma 5.3, immediately gives (5.1). To obtain (5.10), test the Dirichlet quotient on $B^n$ with $1 - |x|^2$. With $\omega_n = |B^n|$, its two integrals are

$$\int_{B^n} |\nabla(1-|x|^2)|^2 \, \mathrm{d}x = \frac{4n\omega_n}{n+2}, \qquad \int_{B^n} (1-|x|^2)^2 \, \mathrm{d}x = \frac{8\omega_n}{(n+2)(n+4)}.$$

The trial function is not a Dirichlet eigenfunction: its negative Laplacian is the constant $2n$, which cannot equal a positive eigenvalue times $1 - |x|^2$. Thus the variational inequality is strict, and

$$\alpha_n^2 < \frac{n(n+4)}{2}, \qquad \frac{\alpha_n^2}{\beta_{n-1}^2} < \frac{n+4}{2},$$

where the second inequality uses Lemma 5.1 with $m = n-1$. Thus (5.10) implies (5.9), in fact strictly. □

*Remark* 5.5 (A further elementary moment bound). Concavity also gives the explicit estimate

$$\frac{1}{|\Omega|}\int_\Omega z^2 \, \mathrm{d}x \geq \frac{2L^2}{(n+1)(n+2)}. \tag{5.11}$$

Indeed, $r(z)/(L-z)$ is nondecreasing on $(0,L)$ by the tangent-line inequality used above. Relative to the probability measure proportional to $(L-z)^{n-1}\,\mathrm{d}z$ on $(0,L)$, both $q(z)=(r(z)/(L-z))^{n-1}$ and $z^2$ are nondecreasing. Their covariance is nonnegative by the argument of Corollary 4.3. All the required moments are integrable because $q(z)(L-z)^{n-1}=r(z)^{n-1}$ and $z^2$ is bounded. It follows that

$$\frac{\int_0^L z^2 r(z)^{n-1}\,\mathrm{d}z}{\int_0^L r(z)^{n-1}\,\mathrm{d}z} \geq \frac{\int_0^L z^2(L-z)^{n-1}\,\mathrm{d}z}{\int_0^L (L-z)^{n-1}\,\mathrm{d}z} = \frac{2L^2}{(n+1)(n+2)}.$$

Consequently another sufficient condition is

$$\frac{L}{R} \geq \sqrt{\frac{(n+1)(n+2)}{2n}}. \tag{5.12}$$

Under this condition, (5.11) and $\beta_{n-1}^2 > n$ give

$$M_{\|} \geq \frac{2L^2}{(n+1)(n+2)} \geq \frac{R^2}{n} > \frac{R^2}{\beta_{n-1}^2}.$$

Proposition 5.2 therefore applies, including at the endpoint of (5.12). For $n>2$, (5.12) is slightly less restrictive than (5.10); at $n=2$ they coincide. Both avoid evaluating Bessel zeros.

### 5.4. **Completion of the main theorems.**

*Proof of Theorems 1.1 and 1.2.* Proposition 3.2 proves all spectral and multiplicity statements in Theorem 1.1. Propositions 3.3 and 4.1 prove its monotonicity and extremum statements, including the assertions on the two pure subspaces at a tie. Every linear combination within the transverse subspace is a positive multiple of $u_e$ for a suitable unit vector $e$, by (3.4). Thus the transverse assertion applies to every first eigenfunction when that branch is strictly lower. The proved positive directional derivatives exclude interior critical points. An interior extremum would be a critical point, so both extrema on the compact closure occur on the boundary and every interior value lies strictly between them. This is (1.1).

For Theorem 1.2, the two axial conditions follow from Proposition 5.2 and Proposition 5.4, followed by Theorem 3.4. The elementary axial condition is (5.12). Its right-hand side is $\sqrt{10/3}<2$ at $n=3$ and $\sqrt{15/4}<2$ at $n=4$. The transverse conditions are Proposition 4.2 and Corollary 4.3. The strict trial inequalities in these arguments justify all endpoint cases. □

## 6. Balls, axial stretching, and examples with unbounded ratio

### 6.1. **The ball, including its full multiple eigenspace.**

**Corollary 6.1.** *Every nonzero first Neumann eigenfunction on $B_R^n$ has a strictly positive derivative in some fixed direction. Its maximum and minimum on $\overline{B_R^n}$ are unique and occur at the two endpoints of the diameter in that direction. This statement includes every linear combination in the first eigenspace, whose dimension is $n$.*

*Proof.* The ball satisfies (G1)–(G3), and its orthogonal symmetries interchange the axial and transverse coordinates. Hence $\lambda_{\|}=\lambda_{\perp}$, and Proposition 3.2 gives $\dim E_1 = n$.

Given $0\neq u\in E_1$, rotate its nonzero vector $J(u)$ to a positive multiple of the axial unit vector. The rotated eigenfunction belongs to the axial line: Proposition 3.2 and injectivity of $J$ eliminate all its transverse components. Proposition 3.3 applies at this tie and gives $J(U)\cdot e_z = \int_{B_R^n} U_z > 0$, so the axial coefficient is positive. Rotating its positive derivative and its two extrema back proves the assertion for $u$. □

6.2. **Only finitely many possible exceptional stretches.**

**Proposition 6.2.** *Fix a body $\Omega_0$ satisfying* (G1)–(G3)*, with axial half-length $L_0$ and equatorial radius $R_0$. For $t > 0$, set*

$$\Omega_t = \{(y, tz) : (y, z) \in \Omega_0\}.$$

*There is a finite nonempty set $\Sigma \subset (0, \infty)$ such that*

$$\lambda_{\|}(\Omega_t) = \lambda_{\perp}(\Omega_t) \quad \Longleftrightarrow \quad t \in \Sigma.$$

*For every $t \notin \Sigma$, every nonzero first eigenfunction on $\Omega_t$ has strict hot spots, unique maximum and minimum, and no interior critical point. The transverse conclusion holds for all sufficiently small $t$, and the axial conclusion for all sufficiently large $t$. No assertion about general mixed eigenfunctions at the values in $\Sigma$ is made, apart from the ball case above.*

*Proof.* The stretched profile is $r_t(z) = r_0(z/t)$ on $(-tL_0, tL_0)$; the stretching is an invertible linear map. It preserves the stated smoothness, symmetry, concavity, and strict profile decrease.

Pull functions back to the fixed domain $\Omega_0$. The Jacobian factor $t$ cancels from each Rayleigh quotient, leaving the form

$$q_t(f, g) = \int_{\Omega_0} (\nabla_y f \cdot \nabla_y g + t^{-2} f_z g_z). \tag{6.1}$$

In each of the two fixed coordinate-odd subspaces of $H^1(\Omega_0)$, its lowest eigenvalue is simple by Lemma 3.1, applied on $\Omega_t$ and pulled back by the same linear map. This remains true at a coincidence between the two sectors, because simplicity is asserted within each fixed odd sector. Both lowest eigenvalues are real analytic in $t > 0$.

Here is a form-level justification of this last assertion. Let $V$ be one of those closed odd subspaces, equipped with its $H^1$ norm. The Poincaré inequality makes $q_t$ coercive on $V$ for each fixed $t > 0$, and the embedding of $V$ into $L^2$ is compact. Let $A_t : V \to V^*$ denote the operator of the bilinear form (6.1), and let $B : V \to V^*$ denote the $L^2$ pairing. The map

$$(t, f, \lambda) \longmapsto \left( A_t f - \lambda B f, \ \frac{\|f\|_2^2 - 1}{2} \right) \quad \text{from } (0, \infty) \times V \times \mathbb{R} \text{ to } V^* \times \mathbb{R}$$

is analytic, since $A_t$ depends analytically on $t$ in operator norm. Fix $t_0 > 0$ and a normalized lowest eigenpair $(f_0, \lambda_0)$ at $t_0$. The derivative in $(f, \lambda)$ is

$$(h, s) \longmapsto \big((A_{t_0} - \lambda_0 B)h - sBf_0, \ (f_0, h)_{L^2}\big).$$

This is an isomorphism. On the $L^2$-orthogonal complement of $f_0$, write $\lambda_+ > \lambda_0$ for the next eigenvalue in $V$. The min–max principle at $t_0$ gives

$$q_{t_0}(h, h) - \lambda_0 \|h\|_2^2 \geq \left(1 - \frac{\lambda_0}{\lambda_+}\right) q_{t_0}(h, h).$$

Uniform ellipticity of $q_{t_0}$ and the Poincaré inequality on the odd subspace make the right-hand side coercive in $H^1$. More explicitly, for a target $(\ell, b) \in V^* \times \mathbb{R}$, write $h = bf_0 + h_\perp$ with $(h_\perp, f_0)_{L^2} = 0$. Testing the first component against $f_0$ forces $s = -\ell(f_0)$. The Lax–Milgram theorem on this closed orthogonal complement gives a unique $h_\perp$ satisfying

$$q_{t_0}(h_\perp, \varphi) - \lambda_0 (h_\perp, \varphi)_{L^2} = \ell(\varphi) \qquad (\varphi \in V, \ (\varphi, f_0)_{L^2} = 0).$$

Together with the equation tested against $f_0$, this solves the full derivative equation. Coercivity gives the bounded inverse. The analytic implicit function theorem supplies an analytic local eigenpair. For all $f \in V$,

$$\min\{1, (t_0/t)^2\}\, q_{t_0}(f, f) \leq q_t(f, f) \leq \max\{1, (t_0/t)^2\}\, q_{t_0}(f, f).$$

As $t \to t_0$, these comparison factors tend to one; min–max therefore gives continuity of the ordered eigenvalues and preserves the gap around $\lambda_0$. This identifies the local eigenvalue with the lowest odd eigenvalue. The local eigenvalue branches agree on overlaps, proving analyticity of the two eigenvalues throughout $(0, \infty)$.

Let $M_{\parallel,0} > 0$ and $M_{\perp,0} > 0$ be the two second moments of $\Omega_0$. Change of variables gives

$$M_{\parallel}(\Omega_t) = t^2 M_{\parallel,0}, \qquad M_{\perp}(\Omega_t) = M_{\perp,0}.$$

The axial half-length is $tL_0$, while the equatorial radius remains $R_0$. For small $t$ the transverse moment condition $M_{\perp,0} \geq 4t^2 L_0^2/\pi^2$ holds, so Proposition 4.2 selects the transverse branch. For large $t$ the axial condition $t^2 M_{\parallel,0} \geq R_0^2/\beta_{n-1}^2$ holds, so Proposition 5.2 and Theorem 3.4 select the axial branch. Consequently the analytic function $\lambda_{\parallel}(\Omega_t) - \lambda_{\perp}(\Omega_t)$ is positive for all sufficiently small $t$ and negative for all sufficiently large $t$. Its zero set is nonempty by continuity and is contained in a compact subinterval of $(0, \infty)$. A nonzero real analytic function has isolated zeros; infinitely many zeros in that compact interval would have an accumulation point. Hence the zero set $\Sigma$ is finite. Theorem 1.1 gives all claimed eigenfunction conclusions off $\Sigma$. □

After pullback to $\Omega_0$, every first eigenfunction with $t \notin \Sigma$ is either odd or even in $z$. These two parity subspaces are closed in $L^2(\Omega_0)$, as is their union. A mixture at a tie with nonzero axial and transverse parts belongs to neither subspace, so it cannot be an $L^2$ limit of these pure first modes. Thus the off-tie conclusions alone do not imply a hot-spots assertion for such mixtures.

**Corollary 6.3.** *For every fixed $\Omega_0$ as in Proposition 6.2, all sufficiently large axial stretches have the strict hot-spots property, while*

$$\frac{d(\Omega_t)^n}{|\Omega_t|} \longrightarrow \infty \qquad (t \longrightarrow \infty).$$

*Proof.* The volume is $|\Omega_t| = t|\Omega_0|$, and the two axial poles are distance $2tL_0$ apart. Therefore

$$\frac{d(\Omega_t)^n}{|\Omega_t|} \geq \frac{2^n L_0^n}{|\Omega_0|}\, t^{n-1} \longrightarrow \infty.$$

The axial conclusion for large $t$ was proved above. □

### 6.3. **An explicit smooth nonellipsoidal family.**

**Proposition 6.4** (Quartic bodies)**.** *Fix $n \geq 2$, $R > 0$, and $0 < c < 1$. For every $L > 0$, the domains (1.9) are nonellipsoidal and satisfy* (G1)–(G3) *with strictly positive principal curvatures. They satisfy all axial hot-spots conclusions whenever*

$$\frac{L}{R} \geq \sqrt{\frac{n+2}{n}}. \tag{6.2}$$

*Their volumes are*

$$|\Omega_{L,R,c}| = \omega_{n-1} L R^{n-1} I_{n,c}, \qquad I_{n,c} = \int_{-1}^{1} [(1-s^2)(1+cs^2)]^{(n-1)/2}\, \mathrm{d}s. \tag{6.3}$$

*If $L/R \geq \sqrt{1+c}$, their diameter is $2L$ and their exact diameter–volume ratio is (1.10). In particular, every sufficiently large $L$ has the strict hot-spots property and the ratio tends to infinity as $L \to \infty$.*

*Proof.* Let

$$F(y,z) = \frac{|y|^2}{R^2} + (1-c)\frac{z^2}{L^2} + c\frac{z^4}{L^4}.$$

Its Hessian is positive definite:

$$D^2F = \operatorname{diag}\left(\frac{2}{R^2}, \dots, \frac{2}{R^2}, \frac{2(1-c)}{L^2} + \frac{12cz^2}{L^4}\right).$$

Its gradient vanishes only at the origin, where $F = 0$, and $F(y,z) \geq |y|^2/R^2 + (1-c)z^2/L^2$ tends to infinity as $|(y,z)| \to \infty$. Hence the level $F = 1$ is a smooth compact hypersurface and bounds the convex domain $\{F < 1\}$. Its outward normal is $\nabla F/|\nabla F|$, so for a tangent vector $T$ on this level,

$$\mathrm{II}(T,T) = \frac{D^2F(T,T)}{|\nabla F|} > 0 \qquad (T \neq 0),$$

which proves positivity of all principal curvatures.

Writing $s = z/L$, its profile is

$$r(z) = R\sqrt{1-(1-c)s^2 - cs^4} = R\sqrt{(1-s^2)(1+cs^2)}. \tag{6.4}$$

Write $A(s) = 1-(1-c)s^2 - cs^4$. It is positive on $(-1,1)$, $A'(s) < 0$ for $0 < s < 1$, and $A''(s) = -2(1-c) - 12cs^2 < 0$ on $(-1,1)$. Since

$$(\sqrt{A})'' = \frac{A''}{2\sqrt{A}} - \frac{(A')^2}{4A^{3/2}} < 0,$$

the profile satisfies $r'(z) < 0$ for $z > 0$ and $r''(z) < 0$ in $(-L, L)$. This verifies (G1)–(G3) explicitly, including the allowed square-root profile at each pole.

These domains are not ellipsoids. The unique center of an ellipsoid with these coordinate reflections would be fixed by each reflection, hence would be the origin. Those reflections eliminate all mixed quadratic terms, and transverse orthogonal symmetry makes the transverse coefficients equal. Its equation would therefore be $a|y|^2 + bz^2 < 1$, so its squared profile would be affine in $z^2$. The nonzero quartic coefficient in (6.4) makes that impossible.

Fubini and the substitution $z = Ls$ give (6.3). The axial second moment is

$$M_{\|} = L^2 \frac{\int_{-1}^{1} s^2(1-s^2)^{(n-1)/2}(1+cs^2)^{(n-1)/2}\,\mathrm{d}s}{\int_{-1}^{1}(1-s^2)^{(n-1)/2}(1+cs^2)^{(n-1)/2}\,\mathrm{d}s}.$$

With respect to the probability density proportional to $(1-s^2)^{(n-1)/2}$, the functions $s^2$ and $(1+cs^2)^{(n-1)/2}$ have strictly positive covariance by the two-point identity used in Corollary 4.3. Indeed, the second function is a strictly increasing function of $s^2$ because $c > 0$ and $n \geq 2$, and pairs with different values of $s^2$ have positive product measure. The untilted second moment is $1/(n+2)$, since

$$0 = \int_{-1}^{1} \left[s(1-s^2)^{(n+1)/2}\right]' \mathrm{d}s = \int_{-1}^{1} (1-(n+2)s^2)(1-s^2)^{(n-1)/2}\,\mathrm{d}s.$$

Consequently

$$M_{\|} > \frac{L^2}{n+2}. \tag{6.5}$$

Under (6.2), Lemma 5.1 therefore gives

$$M_{\|} > \frac{L^2}{n+2} \geq \frac{R^2}{n} > \frac{R^2}{\beta_{n-1}^2}.$$

The axial moment criterion applies even at the stated endpoint.

Finally, apply the diameter calculation to the centrally symmetric compact set $\overline{\Omega_{L,R,c}}$, which has the same diameter as $\Omega_{L,R,c}$. Its diameter is twice the maximum Euclidean norm: the triangle

inequality proves the upper bound and an antipodal pair attains it. The maximum norm occurs on the boundary, since moving outward along a ray increases the norm. There, with $q = s^2 \in [0,1]$,

$$|(y,z)|^2 = R^2 + (L^2 - (1-c)R^2)q - cR^2q^2.$$

The derivative of this polynomial is bounded below by $L^2 - (1+c)R^2$. If $L/R \geq \sqrt{1+c}$, it is therefore nonnegative on $[0,1]$, and the maximum is its value $L^2$ at $q = 1$. Thus $d(\Omega_{L,R,c}) = 2L$. Combining this with (6.3) proves the exact ratio and its divergence. □

**Corollary 6.5** (A single explicit family in every dimension)**.** *For every $n \geq 2$, $R > 0$, and $L/R \geq \sqrt{2}$, the body*

$$\left\{(y,z) : \frac{|y|^2}{R^2} + \frac{z^2}{2L^2} + \frac{z^4}{2L^4} < 1\right\}$$

*has all the axial conclusions above, has diameter $2L$, and satisfies*

$$\frac{d(\Omega)^n}{|\Omega|} = \frac{2^n}{\omega_{n-1} I_{n,1/2}} \left(\frac{L}{R}\right)^{n-1}.$$

*Proof.* Set $c = 1/2$ in Proposition 6.4. For $n \geq 2$, one has $(n+2)/n \leq 2$ and $1 + \frac{1}{2} < 2$. The two conditions in Proposition 6.4 therefore hold. □

6.4. **Exact constants in dimension three.** For $n = 3$ and $c = 1/2$, the relevant integrals are elementary:

$$I_{3,1/2} = \int_{-1}^{1} \left(1 - \frac{s^2}{2} - \frac{s^4}{2}\right) \, \mathrm{d}s = \frac{22}{15}, \qquad \int_{-1}^{1} s^2 \left(1 - \frac{s^2}{2} - \frac{s^4}{2}\right) \, \mathrm{d}s = \frac{34}{105}.$$

Thus the smooth positively curved nonellipsoidal domain

$$\Omega_{L,R,1/2} = \left\{(y_1, y_2, z) : \frac{y_1^2 + y_2^2}{R^2} + \frac{z^2}{2L^2} + \frac{z^4}{2L^4} < 1\right\}$$

has

$$|\Omega_{L,R,1/2}| = \frac{22\pi}{15} LR^2, \qquad M_{\|} = \frac{17}{77} L^2. \tag{6.6}$$

Already the elementary condition $L/R \geq 5/4$ guarantees both axial spectral selection and diameter $2L$. In fact,

$$\frac{17}{77}\left(\frac{5}{4}\right)^2 - \frac{1}{3} = \frac{43}{3696} > 0, \qquad \left(\frac{5}{4}\right)^2 - \frac{3}{2} = \frac{1}{16} > 0.$$

The first inequality and $\beta_2^2 > 3$ imply the axial moment criterion; the second implies the exact diameter condition. For every $L/R \geq 5/4$, therefore, every first eigenfunction has its unique extrema at the axial poles and

$$\boxed{\frac{d(\Omega_{L,R,1/2})^3}{|\Omega_{L,R,1/2}|} = \frac{60}{11\pi}\left(\frac{L}{R}\right)^2 \longrightarrow \infty.} \tag{6.7}$$

This family requires no smallness of a perturbation of the sphere.

## 7. A profile criterion for spectral ordering

The quantity $\mathcal{F}$ in (1.8) measures the change in the distance of the meridional boundary from the origin. Its sign gives a spectral comparison without estimating either eigenvalue separately. We first record the identity used to exclude a coincidence.

**Lemma 7.1** (A polarized rotational identity)**.** *Let $D \subset \mathbb{R}^n$ be a bounded smooth domain, and let smooth real Neumann eigenfunctions $U, V$ have the same eigenvalue $\lambda$:*

$$-\Delta U = \lambda U, \qquad -\Delta V = \lambda V, \qquad \partial_\nu U = \partial_\nu V = 0.$$

*For every skew-symmetric matrix $A$, one has*

$$\int_{\partial D} (Ax \cdot \nu)(\nabla U \cdot \nabla V - \lambda UV)\, \mathrm{d}S = 0. \tag{7.1}$$

*Proof.* Put $K = \nabla U \cdot \nabla V - \lambda UV$ and define the symmetric tensor

$$S_{ij} = K\delta_{ij} - \partial_i U\, \partial_j V - \partial_i V\, \partial_j U.$$

Direct differentiation gives

$$\sum_j \partial_j S_{ij} = -(\Delta U + \lambda U)\partial_i V - (\Delta V + \lambda V)\partial_i U = 0.$$

For $X = Ax$, the contraction of $S$ with $DX$ vanishes because $S$ is symmetric and $A$ is skew-symmetric. The divergence theorem therefore gives

$$\int_{\partial D} X \cdot S\nu \, \mathrm{d}S = 0.$$

The Neumann conditions imply $S\nu = K\nu$, which proves (7.1). □

*Proof of Theorem 1.3.* Suppose first that $\lambda_\parallel = \lambda_\perp = \lambda$. By Proposition 3.2, this is the first positive Neumann eigenvalue. The axial and transverse modes can be written as

$$U = U(\rho, z), \qquad V = v(\rho, z)Y, \qquad Y = \frac{y_1}{\rho},$$

with $U > 0$ for $z > 0$ and $V > 0$ for $y_1 > 0$ in the interior. The axial mode is odd in $z$. The transverse mode may be chosen even in $z$: replace $V$ by $[V(y, z) + V(y, -z)]/2$. Reflection preserves its eigenvalue, Neumann condition, and oddness in $y_1$; positivity of both summands on $y_1 > 0$ makes the average nonzero. The representation and monotonicity results therefore apply to this choice, for which $v$ is even in $z$.

Propositions 3.3 and 4.1 apply also at a coincidence and give

$$U_\rho < 0, \qquad U_z > 0, \qquad v_\rho > 0, \qquad v_z < 0$$

in the upper meridional interior. The eigenfunctions are smooth in Cartesian coordinates up to the axis and the poles. For $n = 2$, the transverse representation means $V(y_1, z) = v(|y_1|, z)\,\mathrm{sgn}(y_1)$.

On the curved upper boundary with $0 < z < L$, both $U$ and $v$ are strictly positive. Indeed, a zero of $U$ there would contradict the Hopf boundary point lemma for its positive upper-half ground state and the Neumann condition. For $v$, apply the same argument to $V$ at $y = r(z)e_1$, away from its Dirichlet cut $y_1 = 0$. The derivative signs extend nonstrictly to this boundary by continuity. Consequently

$$C(z) := (U_\rho v_\rho + U_z v_z - \lambda U v)\,(r(z), z) < 0, \qquad 0 < z < L. \tag{7.2}$$

Apply Lemma 7.1 with $X = ze_1 - y_1 e_z$. On $\rho = r(z)$, the normal and the rotation field satisfy

$$\nu = \frac{e_\rho - r'(z)e_z}{\sqrt{1 + r'(z)^2}}, \qquad X \cdot \nu = \frac{Y\mathcal{F}(z)}{\sqrt{1 + r'(z)^2}}.$$

Since $\nabla U$ has no angular component,

$$\nabla U \cdot \nabla V - \lambda UV = YC(z).$$

Thus the rotational identity becomes

$$0 = \int_{\partial\Omega} \frac{Y^2 \mathcal{F}(z) C(z)}{\sqrt{1 + r'(z)^2}} \, \mathrm{d}S. \tag{7.3}$$

Both $\mathcal{F}$ and $C$ are odd in $z$, so the upper and lower halves make equal contributions.

For the explicit weight, set

$$A(z) = U(r(z), z), \qquad B(z) = v(r(z), z), \qquad 0 < z < L.$$

The Neumann relations give $U_\rho = r' U_z$ and $v_\rho = r' v_z$ on this boundary. Hence

$$A' = (1 + r'^2) U_z, \qquad B' = (1 + r'^2) v_z.$$

Define

$$W(z) = r(z)^{n-2} \left( \lambda A(z) B(z) - \frac{A'(z) B'(z)}{1 + r'(z)^2} \right) = -r(z)^{n-2} C(z) > 0. \tag{7.4}$$

Using

$$\mathrm{d}S = r(z)^{n-2} \sqrt{1 + r'(z)^2} \, \mathrm{d}\theta \, \mathrm{d}z, \qquad \int_{S^{n-2}} Y^2 \, \mathrm{d}\theta = \frac{|S^{n-2}|}{n-1},$$

equation (7.3) reads

$$0 = -\frac{2|S^{n-2}|}{n-1} \int_0^L \mathcal{F}(z) W(z) \, \mathrm{d}z.$$

In particular,

$$\int_0^L \mathcal{F}(z) W(z) \, \mathrm{d}z = 0. \tag{7.5}$$

For $n = 2$, angular integration is over $S^0 = \{-1, 1\}$ and the same formulas apply. All computations originate from the smooth Cartesian boundary identity. The poles and equatorial section have surface measure zero, and absolute convergence in (7.5) follows by applying the same parametrization to the absolute value of its smooth Cartesian boundary integrand. Thus a possibly unbounded graph derivative at a pole creates no additional boundary term.

Since $W > 0$, a continuous $\mathcal{F}$ of one sign and not identically zero cannot satisfy (7.5). This excludes a coincidence under either of the one-sided sign hypotheses. It remains to determine which strict ordering holds.

For the axial stretches $\Omega_t = \{(y, tz) : (y, z) \in \Omega\}$, write $r_t(ts) = r(s)$. Their corresponding quantities satisfy

$$\mathcal{F}_t(ts) = \frac{\mathcal{F}(s) + (t^2 - 1)s}{t}. \tag{7.6}$$

If $\mathcal{F} \geq 0$ and $\mathcal{F} \not\equiv 0$, then $\mathcal{F}_t \geq 0$ and $\mathcal{F}_t \not\equiv 0$ for every $t \geq 1$. The preceding argument rules out a coincidence for every such $t$. Proposition 6.2 gives continuous eigenvalue branches and the strict axial ordering for all sufficiently large $t$. Their continuous difference cannot change sign on $[1, \infty)$ without vanishing. Therefore $\lambda_\parallel(\Omega) < \lambda_\perp(\Omega)$. If $\mathcal{F} \leq 0$ and $\mathcal{F} \not\equiv 0$, then $\mathcal{F}_t \leq 0$ and $\mathcal{F}_t \not\equiv 0$ for $0 < t \leq 1$. The same argument, using the strict transverse ordering for sufficiently small $t$, gives $\lambda_\parallel(\Omega) > \lambda_\perp(\Omega)$.

If $\mathcal{F} \equiv 0$, then $(r(z)^2 + z^2)' = 0$. Continuity at $z = 0$ and $z = L$ gives

$$r(z)^2 + z^2 = R^2, \qquad L = R.$$

The domain is a ball, and its coordinate-interchanging symmetries give $\lambda_\parallel = \lambda_\perp$. Finally, a coincidence for a nonball domain can satisfy neither one-sided sign hypothesis nor $\mathcal{F} \equiv 0$. Thus

$\mathcal{F}$ must take both positive and negative values. The formulas (7.4) and (7.5) also give a necessary balance at every coincidence. □

**Corollary 7.2** (Ellipsoids of revolution)**.** *For the profile* $r(z) = R\sqrt{1 - z^2/L^2}$*, one has*

$$\begin{cases} \lambda_\parallel < \lambda_\perp, & L > R, \\ \lambda_\parallel > \lambda_\perp, & L < R, \\ \lambda_\parallel = \lambda_\perp, & L = R. \end{cases}$$

*Proof.* Direct differentiation gives $\mathcal{F}(z) = z(1 - R^2/L^2)$. The three cases follow from Theorem 1.3. □

### 7.1. **The linear profile: rhombi and double cones.**

**Example 7.3** (A linear meridian profile)**.** Let $L, R > 0$ and consider

$$r_0(z) = R\left(1 - \frac{|z|}{L}\right), \qquad \mathcal{C}_{L,R} = \left\{(y, z) : \frac{|y|}{R} + \frac{|z|}{L} < 1\right\}. \tag{7.7}$$

For $n = 2$ this is a rhombus, and for $n = 3$ it is a circular double cone. It is a convex Lipschitz domain, but the equatorial ridge and the conical poles violate the smoothness assumptions (G1)–(G3). We retain the odd-sector Rayleigh definitions of $\lambda_\parallel$ and $\lambda_\perp$ on this domain. The following calculations give direct sector comparisons and explain how the smooth results apply to rounded versions of it.

Slicing and integration give

$$\begin{aligned} |\mathcal{C}_{L,R}| &= \frac{2\omega_{n-1}}{n} L R^{n-1}, \qquad d(\mathcal{C}_{L,R}) = 2\max\{L, R\}, \\ M_\parallel &= \frac{2L^2}{(n+1)(n+2)}, \qquad M_\perp = \frac{nR^2}{(n+1)(n+2)}. \end{aligned} \tag{7.8}$$

In particular, with $a = L/R$,

$$\frac{d(\mathcal{C}_{L,R})^n}{|\mathcal{C}_{L,R}|} = \frac{n2^{n-1}}{\omega_{n-1}} \frac{\max\{a, 1\}^n}{a}. \tag{7.9}$$

This diverges both as $a \to \infty$ and as $a \downarrow 0$.

The sector eigenvalues satisfy the strict estimates

$$\begin{aligned} \frac{\pi^2}{4L^2} < \lambda_\parallel &< \min\left\{\frac{\alpha_n^2}{L^2}, \frac{(n+1)(n+2)}{2L^2}\right\}, \\ \frac{\beta_{n-1}^2}{R^2} < \lambda_\perp &< \frac{(n+1)(n+2)}{nR^2}. \end{aligned} \tag{7.10}$$

Consequently,

$$a \geq \frac{\alpha_n}{\beta_{n-1}} \quad \Longrightarrow \quad \lambda_\parallel < \lambda_\perp, \tag{7.11}$$

$$a \leq \frac{\pi}{2}\sqrt{\frac{n}{(n+1)(n+2)}} \quad \Longrightarrow \quad \lambda_\parallel > \lambda_\perp. \tag{7.12}$$

Both endpoint ratios are included. For $n = 3$, these conditions read $a \geq \pi/\beta_2$ and $a \leq \pi\sqrt{3/80}$, respectively. The eigenvalues depend continuously on $a$: after scaling $R = 1$ and pulling back to $\mathcal{C}_{1,1}$, their Rayleigh forms have numerator $\int(|\nabla_y u|^2 + a^{-2}|u_z|^2)$ and fixed denominator $\int u^2$. Thus at least one coincidence occurs strictly between the two thresholds. In dimension three, Proposition 7.4 below sharpens these bounds, and Theorem 7.5 proves a unique coincidence and

determines the ordering on either side. In higher dimensions, the displayed estimates alone do not determine the coincidence set.

For $n = 2$ there is an exact comparison for every aspect ratio:

$$\begin{cases} \lambda_{\parallel} < \lambda_{\perp}, & L > R, \\ \lambda_{\parallel} > \lambda_{\perp}, & L < R, \\ \lambda_{\parallel} = \lambda_{\perp}, & L = R. \end{cases} \tag{7.13}$$

Indeed, simplicity and positivity of the mixed ground state on each half-rhombus make it even under the remaining coordinate reflection. Restricting to the quarter-triangle $T = \{y, z > 0 : y/R + z/L < 1\}$ therefore identifies the axial sector with Dirichlet condition on the leg $z = 0$, of length $R$, and Neumann condition on the other two sides. The transverse sector has Dirichlet condition on the leg $y = 0$, of length $L$. Siudeja's strict comparison for the shorter and longer legs of any nonisosceles right triangle [7, Section 3.1, pp. 6–8] gives (7.13); when $L = R$, coordinate interchange gives equality. In that case the rhombus is a square of side $\sqrt{2}R$, and

$$\begin{aligned} \lambda_{\parallel} = \lambda_{\perp} = \mu_1 = \frac{\pi^2}{2R^2}, \\ U(y,z) = \cos\frac{\pi y}{2R}\sin\frac{\pi z}{2R}, \qquad V(y,z) = \sin\frac{\pi y}{2R}\cos\frac{\pi z}{2R}. \end{aligned} \tag{7.14}$$

The mixture $U + V = \sin(\pi(y + z)/(2R))$ attains its maximum along the entire side $y + z = R$. Thus uniqueness of the extreme points in the smooth conclusions cannot be transferred to this nonsmooth limit. For $n \geq 3$, the condition $L = R$ supplies no symmetry interchanging the axial and transverse directions, so it does not by itself give an equality case.

The profile indicator also has a useful limitation in this example. On the open upper meridian,

$$\mathcal{F}_0(z) = z + r_0(z)r_0'(z) = \left(1 + \frac{R^2}{L^2}\right)z - \frac{R^2}{L}, \qquad z_* := \frac{LR^2}{L^2 + R^2}. \tag{7.15}$$

It is negative for $0 < z < z_*$ and positive for $z_* < z < L$, for every $a > 0$. Hence even the formal one-sided sign test would give no ordering for this family. In particular, a sign change alone cannot identify a coincidence, as the exact planar comparison makes explicit.

For completeness, the bounds in (7.10) require no smooth approximation. For fixed $y$, the axial slice has half-length $L(1 - |y|/R) < L$ almost everywhere. The odd interval Poincaré inequality gives the axial lower bound. For fixed $z$, the transverse slice is a ball of radius $r_0(z) < R$ almost everywhere; its odd Poincaré inequality gives the transverse lower bound. Both inequalities are strict when applied to a nonzero sector minimizer, which exists by compactness on the bounded Lipschitz domain. The trials $z$ and $y_1$ give the reciprocal-moment upper bounds in (7.10), using (7.8). Equality would make a nonzero linear harmonic function solve a positive-eigenvalue equation in the interior, which is impossible.

For the remaining axial bound, take the radial Dirichlet ground-state profile $F$ from Lemma 5.3 and put $g(z) = F(1 - z/L)$ for $0 \leq z \leq L$. Its odd extension is an admissible $H^1$ trial. Since the slice weight is a constant multiple of $(L-z)^{n-1}$, its Rayleigh quotient is exactly $\alpha_n^2/L^2$. If this were the sector minimum, the interior eigenvalue equation would give $-g'' = (\alpha_n^2/L^2)g$. Equation (5.7) would then force $(n-1)g'/(L-z) = 0$, contradicting $g' > 0$ on $(0, L)$. This proves strictness also for the linear profile, where the strict weight comparison used for smooth bodies is unavailable.

To place the example within the hypotheses of the main results, one can round the ridge and poles while preserving both intercepts. For $\varepsilon > 0$, set

$$b_\varepsilon = \sqrt{1+\varepsilon^2} + \varepsilon,$$
$$\Omega_\varepsilon = \left\{ (y,z) : \sqrt{\frac{|y|^2}{R^2} + \varepsilon^2} + \sqrt{\frac{z^2}{L^2} + \varepsilon^2} < b_\varepsilon \right\}. \tag{7.16}$$

The defining function has positive definite Hessian and a nonzero gradient on the boundary. Thus $\Omega_\varepsilon$ is smooth and positively curved, with the required symmetries, equatorial radius $R$ and axial half-length $L$. Its profile is

$$r_\varepsilon(z) = R\sqrt{\left(b_\varepsilon - \sqrt{z^2/L^2 + \varepsilon^2}\right)^2 - \varepsilon^2}. \tag{7.17}$$

These domains satisfy (G1)–(G3), so the sector classification and directional monotonicity results apply to each of them, and either aspect-ratio condition (7.11) or (7.12) gives the corresponding strict hot-spots conclusion by Theorem 1.2. As $\varepsilon \downarrow 0$, $r_\varepsilon$ converges uniformly to $r_0$ on $[-L, L]$, and the volumes and second moments converge to (7.8). Hence any strict margin in a moment condition for the limiting profile also holds for all sufficiently small $\varepsilon$. This applies the smooth theorems to actual smooth domains without assuming that their pointwise conclusions persist at the singular limit. For $n = 2$ and $L = R$, coordinate interchange also gives $\lambda_\parallel(\Omega_\varepsilon) = \lambda_\perp(\Omega_\varepsilon)$ for every $\varepsilon > 0$.

7.2. **The unique transition for the three-dimensional double cone.** The planar equality case $L = R$ is not an equality case for the circular double cone in three dimensions. A positive polynomial on the upper half-cone gives a direct comparison throughout the range $L \le R$. Higher-degree polynomial comparisons extend the transverse ordering to $L > R$ and give a sharper axial condition for long cones. A directional-derivative estimate then shows that every coincidence is a strict downward crossing, which yields a unique transition.

**Proposition 7.4** (Polynomial bounds and localization for circular double cones)**.** *Let $n = 3$ and let $\mathcal{C}_{L,R}$ be the domain (7.7). Write $a = L/R$. For every $L, R > 0$,*

$$\lambda_\perp < \frac{83}{15R^2} + \frac{3}{10L^2}. \tag{7.18}$$

*If $0 < L \le R$, then*

$$\lambda_\perp < \frac{83}{15R^2} + \frac{3}{10L^2} < \frac{4}{R^2} + \frac{2}{L^2} < \lambda_\parallel. \tag{7.19}$$

*In particular, at $L = R$,*

$$\lambda_\perp < \frac{35}{6R^2} < \frac{6}{R^2} < \lambda_\parallel. \tag{7.20}$$

*More generally, the following strict comparisons hold:*

$$0 < a \le \frac{\sqrt{5}}{2} \quad \Longrightarrow \quad \lambda_\parallel > \lambda_\perp, \tag{7.21}$$
$$a \ge \sqrt{\frac{339}{181}} \quad \Longrightarrow \quad \lambda_\parallel < \lambda_\perp. \tag{7.22}$$

*Both threshold endpoints are included. Their approximate values are $\sqrt{5}/2 = 1.1180339\ldots$ and $\sqrt{339/181} = 1.3685496\ldots$. Consequently a coincidence aspect ratio exists in*

$$\frac{\sqrt{5}}{2} < \frac{L}{R} < \sqrt{\frac{339}{181}}, \tag{7.23}$$

*and every coincidence belongs to this interval. Theorem 7.5 below shows that there is exactly one and determines the ordering on both sides.*

*Proof.* We first consider $L \le R$ for the axial comparison; the transverse polynomial trial below is valid for every $L, R > 0$. Put $a = L/R$, $s = \rho/R$, and $t = z/L$. On the upper half-cone $D = \mathcal{C}_{L,R} \cap \{z > 0\}$, consider the Cartesian polynomial

$$w(y, z) = t\left(1 - s^2 - \frac{t^2}{3}\right).$$

It vanishes on the flat base and satisfies $0 < w < t$ in $D$. On the conical side $s + t = 1$, the outward normal is a positive multiple of $e_\rho/R + e_z/L$. Thus $\partial_\nu w$ has the same sign as

$$a^2 w_s + w_t = 1 - s^2 - t^2 - 2a^2 st = 2(1 - a^2)st \ge 0.$$

Moreover,

$$-\Delta w = kt, \qquad k = \frac{4}{R^2} + \frac{2}{L^2}.$$

Let $U > 0$ be the axial mixed ground state on $D$, with Dirichlet condition on its base and Neumann condition on its conical side. The variational equation tested with $w$, followed by Green's formula for the smooth polynomial $w$, gives

$$\lambda_\parallel \int_D Uw = \int_D \nabla U \cdot \nabla w = k \int_D Ut + \int_{\partial D \setminus \{z=0\}} U \partial_\nu w \ge k \int_D Ut > k \int_D Uw.$$

Here the trace of $U$ is nonnegative, the base term is zero, and the ridge and pole have surface measure zero. This is a Lipschitz-domain integration by parts; it requires no smoothness of the cone. Positivity of $U$ and $t - w$ proves the strict axial lower bound, including $a = 1$.

For the transverse bound use the odd polynomial trial

$$P(y, z) = \frac{y_1}{R}\left(1 - \frac{z^2}{L^2} - \frac{\rho^2}{3R^2}\right).$$

Let $T = \{s, t > 0 : s + t < 1\}$ and $v = s(1 - t^2 - s^2/3)$, so that $P = v \cos\theta$. Angular integration and reflection across the equator cancel the common factors in its Rayleigh quotient. The remaining integrals are

$$\int_T sv^2 = \frac{2}{63}, \qquad \int_T s\left(v_s^2 + \frac{v^2}{s^2}\right) = \frac{166}{945}, \qquad \int_T sv_t^2 = \frac{1}{105}.$$

They follow by expanding the polynomials and using $\int_T s^p t^q \, \mathrm{d}s \, \mathrm{d}t = p!q!/(p+q+2)!$. Consequently

$$Q(P) = \frac{R^{-2}(166/945) + L^{-2}(1/105)}{2/63} = \frac{83}{15R^2} + \frac{3}{10L^2}.$$

The variational inequality $\lambda_\perp < Q(P)$ is strict: an equality would make $P$ an interior eigenfunction, whereas

$$-\Delta P = \left(\frac{8}{3R^2} + \frac{2}{L^2}\right)\frac{y_1}{R}$$

cannot be a positive constant times $P$. This proves (7.18) for all $L, R > 0$. Finally, for $L \le R$,

$$k - Q(P) = \frac{17}{10L^2} - \frac{23}{15R^2} \ge \frac{1}{6R^2} > 0 \qquad (L \le R).$$

This proves (7.19) and (7.20).

*The transverse ordering for* $1 \le a \le \sqrt{5}/2$. Uniform scaling by $R$ multiplies both eigenvalues by $R^{-2}$, so normalize $R = 1$, $L = a$, and put $q = a^2 \ge 1$. Retain $s = \rho$ and $t = z/a$, and define

$$F = s^2 + \frac{t^2}{3}, \qquad H = s^4 + 2s^2t^2 + \frac{t^4}{5}.$$

On $\overline{T}$ one has

$$H \le (s+t)^2\left(s^2+\frac{t^2}{5}\right) \le s^2+\frac{t^2}{5}. \tag{7.24}$$

Indeed, the first difference is $\frac{2}{5}st(5s^2-2st+t^2) \ge 0$. The same inequality holds with $s$ and $t$ exchanged.

Set

$$G=\frac{6F-H}{5}, \qquad k_q=\frac{2}{1+q}, \qquad w_q=t(1-k_qG), \qquad \Lambda(q)=\frac{83}{15}+\frac{3}{10q}.$$

Since

$$H-F^2=\frac{4}{3}s^2t^2+\frac{4}{45}t^4 \ge 0, \qquad H \le (s^2+t^2)^2 \le 3F,$$

we have $3F/5 \le G \le (6F-F^2)/5 < 1$ in $T$. Thus $0 < w_q < t$ in the upper half-cone, and $w_q = 0$ on its flat base. On the conical side $s+t=1$, direct differentiation gives

$$q(w_q)_s+(w_q)_t=\frac{q-1}{q+1}(1-4st)\left(1+\frac{4}{5}st\right) \ge 0. \tag{7.25}$$

Consequently $\partial_\nu w_q \ge 0$ there.

For an axisymmetric function, the Cartesian Laplacian becomes $\Delta = \partial_{ss}+s^{-1}\partial_s+q^{-1}\partial_{tt}$. Define $A_q=(-\Delta w_q-\Lambda(q)w_q)/t$. Its polynomial expression is

$$\begin{aligned} A_q = k_q\left[\frac{6}{5}\left(4+\frac{2}{q}\right)-\frac{1}{5}\left(\left(16+\frac{12}{q}\right)s^2+\left(8+\frac{4}{q}\right)t^2\right)\right] \\ -\Lambda(q)(1-k_qG). \end{aligned} \tag{7.26}$$

At the two endpoint values of $q$, inequality (7.24) gives

$$\begin{aligned} 30A_1 &= 41+42s^2-2t^2-35H \ge 41+7s^2-9t^2 \ge 32, \\ 16875A_{5/4} &= 3375+27120s^2+1040t^2-17320H \\ &\ge 3375+9800s^2-2424t^2 \ge 951. \end{aligned}$$

For each fixed $(s,t)$, the expression $q(q+1)A_q$ is a concave quadratic polynomial in $q$, with leading coefficient $-83/15$. Its positive values at $q=1$ and $q=5/4$ therefore give $A_q > 0$ for every $1 \le q \le 5/4$.

Apply the preceding weak Green identity to the positive axial mixed ground state $U$ and the smooth polynomial $w_q$:

$$\lambda_\parallel \int_D Uw_q = \int_D U(-\Delta w_q)+\int_{\partial D\setminus\{z=0\}} U\partial_\nu w_q > \Lambda(q)\int_D Uw_q.$$

The side term is nonnegative by (7.25), and the interior inequality is strict because $tA_q > 0$. The earlier transverse polynomial trial has quotient $\Lambda(q)$ for every $q > 0$, independently of the assumption $L \le R$. Its non-eigenfunction argument gives $\lambda_\perp < \Lambda(q)$. Hence $\lambda_\perp < \lambda_\parallel$ for $1 \le a \le \sqrt{5}/2$. Together with the already proved range $a \le 1$, this proves (7.21).

*The axial ordering for $a \ge \sqrt{339/181}$.* Continue with $R=1$ and $q=a^2 \ge 1$. This time define

$$\widehat{F}=t^2+\frac{s^2}{3}, \qquad \widehat{H}=t^4+2s^2t^2+\frac{s^4}{5}, \qquad g=1-\frac{6}{5}\widehat{F}+\frac{1}{5}\widehat{H}, \qquad P_*(y,z)=y_1g.$$

The identity $\widehat{H} \ge \widehat{F}^2$ gives $g \ge (1-\widehat{F})(1-\widehat{F}/5) > 0$ inside the cone. Thus $P_*$ is positive on the transverse half-domain $D_\perp = \mathcal{C}_{a,1} \cap \{y_1 > 0\}$ and vanishes on its Dirichlet cut. For $v_* = sg$, its conormal derivative on the upper side satisfies

$$q(v_*)_s+(v_*)_t=\frac{8}{5}(q-1)st(1+st) \ge 0. \tag{7.27}$$

Multiplication by $y_1/\rho > 0$ gives the sign of $\partial_\nu P_*$ on this half-domain. Reflection across $z = 0$ gives the same sign on the lower side.

Put $C(q) = 8/3 + 2/q$ and $B(q) = 6C(q)/5 = 16/5 + 12/(5q)$. The full Cartesian Laplacian of $P_*$ satisfies

$$\frac{-\Delta P_*}{y_1} = B(q) - \frac{1}{5}\left[\left(\frac{24}{5} + \frac{4}{q}\right)s^2 + \left(16 + \frac{12}{q}\right)t^2\right].$$

Subtract $B(q)P_*$ and use $\widehat{H} \le t^2 + s^2/5$, the exchanged form of (7.24). The $t^2$ coefficient is zero and the $s^2$ coefficient is positive, giving

$$\frac{-\Delta P_* - B(q)P_*}{y_1} \ge \frac{24 + 8/q}{125}s^2 > 0 \qquad \text{in } D_\perp. \tag{7.28}$$

If $V > 0$ is the transverse mixed ground state, the same Lipschitz-domain Green formula now yields

$$\lambda_\perp \int_{D_\perp} VP_* = \int_{D_\perp} V(-\Delta P_*) + \int_{\partial D_\perp \setminus \{y_1=0\}} V\partial_\nu P_* > B(q)\int_{D_\perp} VP_*.$$

Here the cut term is zero, the conical side term is nonnegative, and (7.28) makes the interior inequality strict. Therefore $\lambda_\perp > B(q)$ for every $q \ge 1$.

For an axial upper bound use the odd polynomial trial

$$u_* = t\left(1 - \frac{4}{5}s^2 - \frac{1}{4}t^2\right).$$

With $\mathbb{E}$ denoting normalized volume average on $\mathcal{C}_{a,1}$, direct integration gives

$$\mathbb{E}u_*^2 = \frac{6763}{100800}, \qquad \mathbb{E}(u_*)_s^2 = \frac{32}{875}, \qquad \mathbb{E}(u_*)_t^2 = \frac{277}{560}.$$

Indeed, after angular integration and equatorial reflection, the normalized measure on $T$ is $6s\ \mathrm{d}s\ \mathrm{d}t$, so these values follow from the monomial integral used above. Consequently

$$\begin{aligned} Q(u_*) &= \frac{18432}{33815} + \frac{49860}{6763q}, \\ B(q) - Q(u_*) &= \frac{496(181q - 339)}{33815q}. \end{aligned} \tag{7.29}$$

For $q \ge 339/181$ it follows that $\lambda_\parallel \le Q(u_*) \le B(q) < \lambda_\perp$. This proves (7.22), including its endpoint. Scaling back restores the common factor $R^{-2}$.

Finally, the two strict comparisons exclude every coincidence outside (7.23). The sector eigenvalues are continuous in $a$: after pullback to $\mathcal{C}_{1,1}$, their Rayleigh numerators have the form $E_y + a^{-2}E_z$, with nonnegative energies and fixed denominator. Comparison at $a$ and $a_0$ bounds each eigenvalue between

$$\min\{1, a_0^2/a^2\}\lambda(a_0) \quad \text{and} \quad \max\{1, a_0^2/a^2\}\lambda(a_0).$$

Thus continuity and the opposite strict signs at the two thresholds give at least one coincidence strictly between them. □

**Theorem 7.5** (Unique equality ratio for circular double cones)**.** *There is a unique universal constant*

$$a_* \in \left(\frac{\sqrt{5}}{2}, \sqrt{\frac{339}{181}}\right) \tag{7.30}$$

*such that the three-dimensional double cone $\mathcal{C}_{L,R}$ satisfies*

$$\begin{cases} \lambda_{\parallel} > \lambda_{\perp}, & L/R < a_*, \\ \lambda_{\parallel} = \lambda_{\perp}, & L/R = a_*, \\ \lambda_{\parallel} < \lambda_{\perp}, & L/R > a_*. \end{cases} \tag{7.31}$$

*For the normalized gap*

$$D(a) = \lambda_{\parallel}(\mathcal{C}_{a,1}) - \lambda_{\perp}(\mathcal{C}_{a,1}),$$

*the equality is a simple crossing, with*

$$D'(a_*) < -\frac{1}{6a_*}. \tag{7.32}$$

The explicit comparisons in Proposition 7.4 locate every coincidence and supply its existence. To prove uniqueness, we show that the eigenvalue gap has a strictly negative derivative at every coincidence. The straight conical side gives a directional estimate for the axial ground state that is stronger than the separate signs of its meridional derivatives.

**Lemma 7.6** (An axial directional estimate)**.** *Let $a > 1$, put $\mathcal{C}_a = \mathcal{C}_{a,1}$ and $D_a = \mathcal{C}_a \cap \{z > 0\}$, and write $A = \lambda_{\parallel}(\mathcal{C}_a)$. Choose the axial odd ground state $U$ positive on $D_a$ and normalized by $\int_{\mathcal{C}_a} U^2 = 1$. Then, almost everywhere on $D_a$,*

$$U_z \geq a|U_\rho|, \qquad \int_{\mathcal{C}_a} |U_z|^2 \geq \frac{a^2}{1+a^2} A. \tag{7.33}$$

*Proof.* Restriction to $D_a$ identifies the axial odd Rayleigh minimum with the first mixed eigenvalue having Dirichlet condition on $z = 0$ and Neumann condition on the conical side. Indeed, the trace of an odd $H^1$ function on the coordinate cut is zero, and odd reflection of an $H^1$ function with zero cut trace belongs to $H^1$ on the full cone; reflection doubles both integrals in the quotient. Compactness on the bounded Lipschitz half-cone attains the minimum. Its value is positive, since a zero-energy minimizer would be a constant with zero cut trace. Taking absolute values, followed by the strong maximum principle, gives a strictly positive ground state. Every real ground-state eigenfunction has constant sign; otherwise its absolute value would be a nonnegative ground state with an interior zero. Two independent ground states would have a nonzero linear combination with zero integral, a contradiction. Thus the ground state is simple. These arguments apply also to the transverse half-cone used below. Odd reflection of the mixed weak equation gives the full homogeneous Neumann equation. Simplicity and rotation invariance make $U$ axially symmetric.

We first justify the derivative signs

$$U_z \geq 0, \qquad U_\rho \leq 0 \quad \text{in } D_a. \tag{7.34}$$

The axial odd-ground-state theorem of Chen–Li–Wang [1, Theorem 3.1] gives the corresponding strict interior signs on smooth strictly simple rotationally symmetric domains: these are equatorially symmetric domains whose upper boundary is a strictly decreasing function of the distance from the axis. That theorem concerns the lowest axial odd eigenfunction itself; its use does not require that the axial mode already belong to the first positive full Neumann eigenspace. We pass from smooth domains to the cone explicitly, so no regularity convention at a conical vertex is needed.

For $0 < \varepsilon < 1/4$, set

$$\mathcal{K}_\varepsilon = \left\{ (y,z) : \sqrt{|y|^2+\varepsilon^2} + \sqrt{z^2/a^2+\varepsilon^2} < 1 \right\}. \tag{7.35}$$

These domains are smooth, strictly convex, rotationally and equatorially symmetric. Their upper meridian is

$$z = f_\varepsilon(\rho) = a\sqrt{\left(1-\sqrt{\rho^2+\varepsilon^2}\right)^2 - \varepsilon^2},$$

which is strictly decreasing wherever $\rho > 0$ and $f_\varepsilon(\rho) > 0$. Thus they are strictly simple in the sense required by the cited theorem. They satisfy

$$(1-2\varepsilon)\mathcal{C}_a \subset \mathcal{K}_\varepsilon \subset \mathcal{C}_a.$$

Their normalized axial odd ground states $U_\varepsilon$ have the required derivative signs by the cited theorem, and their eigenvalues are uniformly bounded by using $z$ as a trial.

Here are the compactness details for this approximation. All $\mathcal{K}_\varepsilon$ contain one fixed centered ball and lie in another. The Minkowski gauges of these convex domains are uniformly Lipschitz and bounded above and below on the unit sphere. Their radial boundary functions therefore have the same uniform Lipschitz and positive lower bounds. The radial maps from the unit ball to $\mathcal{K}_\varepsilon$, extended homogeneously along every ray in $\mathbb{R}^3$, are uniformly bi-Lipschitz. Pull back to the unit ball and extend into $1 < |x| < 3/2$ by the reflection $f(r\omega) = f((2-r)\omega)$. Multiply the extension by a fixed smooth radial cutoff equal to one on the closed unit ball and supported in $|x| < 3/2$. The reflection is bi-Lipschitz between these collars, since its reflected radius stays above $1/2$, so this is a bounded $H^1$ extension. Pushing it forward by the same homogeneous map gives an extension on $\mathbb{R}^3$ with uniform $H^1$ norm and support in one fixed ball. Every operation commutes with $z$-reflection, so the extensions remain odd. A subsequence converges weakly in $H^1$ and strongly in $L^2$ on that ball. The shrinking boundary band has vanishing mass by strong $L^2$ convergence, so its limit has norm one on $\mathcal{C}_a$. Lower semicontinuity on a compact exhaustion of the cone gives the lower bound for its limiting Rayleigh energy. Conversely, approximate a cone ground state in $H^1$ by smooth ambient odd functions and restrict these to $\mathcal{K}_\varepsilon$. Such approximations follow from Lipschitz extension, mollification, and odd symmetrization. The omitted boundary band has vanishing measure, so first letting $\varepsilon \downarrow 0$ and then taking the smooth approximation limit gives the matching upper bound for the eigenvalues. Thus the limit is the normalized positive axial odd cone ground state, by simplicity. Interior elliptic estimates for $-\Delta U_\varepsilon = \lambda_\varepsilon U_\varepsilon$ upgrade the convergence to $C^1$ on compact subsets of the cone. The derivative signs pass to the limit and give (7.34). No derivative at a vertex is asserted.

We also need Sobolev regularity on the actual cone. Mitrea's homogeneous Neumann regularity theorem [6, Corollary 6.6] gives

$$U \in H^2(\mathcal{C}_a). \tag{7.36}$$

Indeed, the full cone is a bounded connected convex Lipschitz domain. It has the uniform exterior ball condition: at each boundary point, a ball of any fixed radius tangent from the exterior supporting half-space remains outside the convex domain. The forcing $AU$ belongs to $L^2$ and has mean zero by oddness. Applying the cited theorem to the full weak Neumann extension avoids any assumption of mixed regularity at the equatorial edge.

On $D_a$ put

$$p = U_z \geq 0, \qquad q = -U_\rho = |\nabla_y U| \geq 0, \qquad H = p - aq = U_z + aU_\rho.$$

The Sobolev chain rule for the Euclidean norm, together with (7.36), gives $q, H \in H^1(D_a)$. For $\rho > 0$, the two eigenvalues of the transverse radial Hessian $D_y^2 U$ are $U_{\rho\rho}$ and $U_\rho/\rho$. Hence

$$\left| D_y^2 U \right|^2 = |U_{\rho\rho}|^2 + |U_\rho/\rho|^2 \quad \text{almost everywhere,}$$

and (7.36) gives $U_\rho/\rho \in L^2(D_a)$. On the slant, the outward normal is proportional to $e_\rho + a^{-1}e_z$, so the Neumann trace gives $H = 0$. On the flat base, the zero trace of $U$ has zero tangential derivative, hence $U_\rho = 0$, while the trace of $p$ is nonnegative. These assertions are trace identities on the open smooth boundary faces: the $H^2$ regularity permits tangential differentiation of the zero Dirichlet trace, and the trace of a nonnegative $H^1$ function is nonnegative. The exceptional ridge and vertex have surface measure zero. Thus the negative part of $H$ has zero trace almost everywhere on the whole boundary. Since $D_a$ is Lipschitz, the kernel of its $H^1$ trace map is $H^1_0(D_a)$. Consequently

$$h := H_- \in H^1_0(D_a), \qquad 0 \le h \le aq, \qquad h/\rho \in L^2(D_a).$$

Away from the rotation axis, differentiating the eigenvalue equation in cylindrical coordinates gives

$$(-\Delta - A)p = 0, \qquad (-\Delta - A)U_\rho = -\frac{U_\rho}{\rho^2}.$$

Here $\Delta = \partial_{\rho\rho} + \rho^{-1}\partial_\rho + \partial_{zz}$ on axially symmetric functions; in particular, the second identity includes the derivative of $\rho^{-1}\partial_\rho$. It follows that

$$(-\Delta - A)H = -\frac{a}{\rho^2}U_\rho \ge 0. \tag{7.37}$$

To justify negative-part testing across the axis, choose radial cutoffs $0 \le \chi_\varepsilon \le 1$, equal to zero for $\rho \le \varepsilon$ and to one for $\rho \ge 2\varepsilon$, with $|\nabla\chi_\varepsilon| \le C/\rho$. Test (7.37) with $\chi_\varepsilon h \in H^1_0(D_a)$. For fixed $\varepsilon$, this is legitimate by approximating the test in $H^1$ by compactly supported smooth tests away from the axis: all coefficients of (7.37) are bounded on the support, and its right-hand side belongs there to $L^2$. Its right-hand side pairs nonnegatively with this test. It is integrable, since $qh/\rho^2 \le aq^2/\rho^2 \in L^1(D_a)$. Writing $E_\varepsilon = D_a \cap \{\varepsilon < \rho < 2\varepsilon\}$, the cutoff error is bounded by

$$C\|\nabla H\|_{L^2(E_\varepsilon)}\|h/\rho\|_{L^2(E_\varepsilon)} \longrightarrow 0.$$

Since $\nabla H \cdot \nabla h = -|\nabla h|^2$ and $Hh = -h^2$ almost everywhere, the tested equation is

$$-\int_{D_a}\chi_\varepsilon|\nabla h|^2 + \int_{D_a} h\nabla H\cdot\nabla\chi_\varepsilon + A\int_{D_a}\chi_\varepsilon h^2 = a\int_{D_a}\chi_\varepsilon\frac{qh}{\rho^2}.$$

All terms except the vanishing cutoff error converge by dominated convergence. Hence the full limiting identity is

$$\int_{D_a}|\nabla h|^2 + a\int_{D_a}\frac{qh}{\rho^2} = A\int_{D_a}h^2,$$

which in particular gives

$$\int_{D_a}|\nabla h|^2 \le A\int_{D_a}h^2. \tag{7.38}$$

The first full Dirichlet eigenvalue of $D_a$ strictly exceeds its mixed ground value $A$. Variational inclusion gives the weak inequality. If equality held, the positive Dirichlet ground state would also be a mixed minimizer, and would have both zero trace and zero normal derivative on an open smooth slant patch. Boundary regularity on that patch and the Hopf boundary lemma give a contradiction. Writing this Dirichlet eigenvalue as $\lambda_1^D(D_a)$, the variational inequality for $h \in H^1_0(D_a)$ and the limiting identity give

$$0 \le (\lambda_1^D(D_a) - A)\int_{D_a}h^2 \le -a\int_{D_a}\frac{qh}{\rho^2} \le 0.$$

Since $\lambda_1^D(D_a) > A$, this forces $h = 0$. Thus $U_z \ge a|U_\rho|$ on $D_a$. Squaring, integrating, and reflecting across $z = 0$ give

$$\int_{\mathcal{C}_a}|U_\rho|^2 \le a^{-2}\int_{\mathcal{C}_a}|U_z|^2.$$

Since $U$ is normalized and axially symmetric, $A = \int_{\mathcal{C}_a}(|U_z|^2 + |U_\rho|^2)$. This proves (7.33). □

*Second proof of Lemma 7.6.* We obtain the derivative signs directly on the cone. The mixed variational characterization, positivity and simplicity of its ground state, and the full-cone $H^2$ regularity (7.36) use no monotonicity theorem. We use these facts, with the same normalization of $U$. Write $\Gamma_0 = \partial D_a \cap \{z = 0\}$ for the base and $\Gamma_1$ for the open conical side, and put

$$X = \nabla U = se_\rho + pe_z, \qquad s = U_\rho, \quad p = U_z, \quad v = s_+, \quad w = p_-.$$

Here $e_\rho = y/|y|$ for $\rho > 0$, viewed as a transverse unit vector in $\mathbb{R}^3$. Since $s$ is signed at this stage, we justify its Sobolev regularity before using truncations. Away from the axis, axial symmetry gives

$$\nabla s = D^2U\, e_\rho, \qquad |D_y^2 U|^2 = |U_{\rho\rho}|^2 + |s/\rho|^2.$$

Thus the candidate weak gradient of $s$ is in $L^2$, as is $s/\rho$. To verify the derivative across the axis, let $\eta_\delta$ vanish for $\rho \le \delta$, equal one for $\rho \ge 2\delta$, and satisfy $|\nabla\eta_\delta| \le C/\rho$. For $\varphi \in C_c^\infty(D_a)$, the error in the distributional derivative identity is bounded by

$$\left|\int_{D_a} s\varphi\nabla\eta_\delta\right| \le C\|s/\rho\|_{L^2(E_\delta)}\|\varphi\|_{L^2(E_\delta)} \longrightarrow 0, \qquad E_\delta = D_a \cap \{\delta < \rho < 2\delta\}.$$

The remaining terms converge by their $L^2$ bounds. Hence $s \in H^1(D_a)$. Also $p \in H^1(D_a)$ by $U \in H^2$, so $v, w \in H^1(D_a)$ by the Sobolev chain rule. On the open base, odd reflection makes $U$ smooth across $z = 0$. Its zero trace gives $s = 0$ there, and positivity for $z > 0$ gives $p \ge 0$ there. Thus both $v$ and $w$ have zero base trace. On $\Gamma_1$, the Neumann condition gives

$$p = -as, \qquad w = av.$$

It follows that the meridional vector field

$$Y = ve_\rho - we_z$$

has zero base trace and is tangent to the conical side. Away from the axis its derivative satisfies

$$|DY|^2 = |\nabla v|^2 + |\nabla w|^2 + \frac{v^2}{\rho^2}.$$

The right-hand side is integrable, since $v \le |s|$. Applying the same axis-cutoff argument to $ve_\rho$, and using $w \in H^1$, proves $Y \in H^1(D_a; \mathbb{R}^3)$ on the whole half-cone.

We justify the following differentiated variational identity without any regularity assumption at the ridge or vertex:

$$\int_{D_a} DX : DY = A\int_{D_a} X \cdot Y. \tag{7.39}$$

Here the colon denotes the Euclidean inner product of matrices. Let $Y_k = \min\{1, k/|Y|\}Y$, with value zero where $Y = 0$. Projection onto a Euclidean ball is 1-Lipschitz; the weak chain rule and locality of weak derivatives give

$$|Y - Y_k| \le |Y|\mathbf{1}_{\{|Y|\ge k\}}, \qquad |D(Y - Y_k)| \le 2|DY|\mathbf{1}_{\{|Y|\ge k\}} \quad \text{almost everywhere.}$$

Thus $Y_k \to Y$ strongly in $H^1$ and $|Y_k| \le k$. The trace map commutes with Lipschitz compositions, so $Y_k$ retains zero base trace and meridional tangency.

We next give explicit cutoffs at both nonsmooth boundary sets. Let $E = \{\rho = 1, z = 0\}$ be the ridge circle, $P = (0, a)$ the vertex, $d(x) = \operatorname{dist}(x, E)$, and $r(x) = |x - P|$. For $0 < \varepsilon < 1/4$, set $\ell_\varepsilon = \log(1/\varepsilon)$ and define

$$\alpha_\varepsilon(x) = \begin{cases} 0, & d(x) \le \varepsilon^2, \\ \log(d(x)/\varepsilon^2)/\ell_\varepsilon, & \varepsilon^2 < d(x) < \varepsilon, \\ 1, & d(x) \ge \varepsilon. \end{cases}$$

Choose a fixed Lipschitz function $\psi : [0,\infty) \to [0,1]$ equal to zero on $[0,1]$ and one on $[2,\infty)$, and put $\beta_\varepsilon(x) = \psi(r(x)/\varepsilon)$ and $\chi_\varepsilon = \alpha_\varepsilon\beta_\varepsilon$. In a fixed tube around $E$, cylindrical coordinates give $d^2 = (\rho-1)^2 + z^2$ and volume element $\rho\,\mathrm{d}\rho\,\mathrm{d}z\,\mathrm{d}\theta$. Polar coordinates in $(\rho-1, z)$ therefore bound the volume element by a constant times $d\,\mathrm{d}d$ after angular integration. Consequently

$$\int_{D_a} |\nabla\alpha_\varepsilon|^2 \le \frac{C}{\ell_\varepsilon^2}\int_{\varepsilon^2}^{\varepsilon} \frac{\mathrm{d}d}{d} = \frac{C}{\ell_\varepsilon}, \qquad \int_{D_a} |\nabla\beta_\varepsilon|^2 \le C\varepsilon^{-2}|B_{2\varepsilon}(P)| \le C\varepsilon.$$

Here $C$ is independent of $\varepsilon$ and $k$. Thus $\chi_\varepsilon$ is Lipschitz, vanishes near $S = E \cup \{P\}$, converges pointwise to one off $S$, and

$$\|\nabla\chi_\varepsilon\|^2_{L^2(D_a)} \le C(\ell_\varepsilon^{-1} + \varepsilon) \longrightarrow 0.$$

The vector equation $-\Delta X = AX$ holds componentwise. For fixed $\varepsilon, k$, the support of $\chi_\varepsilon Y_k$ stays a positive distance from $S$. On this support $U$ is smooth up to each open boundary face. A finite partition into interior and smooth boundary charts, followed by $H^1$ approximation in those charts, therefore justifies the ordinary componentwise Green formula for this $H^1$ test field. Its boundary term is zero on $\Gamma_0$. On $\Gamma_1$, the normal $\nu = (ae_\rho + e_z)/\sqrt{1+a^2}$ is constant along every meridional generator. Since $Y_k$ is tangent to that generator, tangential differentiation of $\partial_\nu U = 0$ gives

$$(D^2U\,\nu)\cdot Y_k = \partial_{Y_k}(\partial_\nu U) - \nabla U\cdot\partial_{Y_k}\nu = 0.$$

The localized Green formula consequently gives

$$\int_{D_a} DX : D(\chi_\varepsilon Y_k) - A\int_{D_a} X\cdot\chi_\varepsilon Y_k = \int_{\Gamma_0\cup\Gamma_1} \chi_\varepsilon (D^2U\,\nu)\cdot Y_k = 0.$$

No axis boundary is introduced: this is Cartesian integration by parts with an $H^1$ test field. Fix $k$ before removing the cutoffs. Dominated convergence gives $(\chi_\varepsilon - 1)Y_k \to 0$ and $(\chi_\varepsilon - 1)DY_k \to 0$ in $L^2$, while

$$\|Y_k \otimes \nabla\chi_\varepsilon\|_{L^2(D_a)} \le k\|\nabla\chi_\varepsilon\|_{L^2(D_a)} \longrightarrow 0.$$

Thus $\chi_\varepsilon Y_k \to Y_k$ in $H^1$. In particular, the cutoff error in the Green formula is bounded by

$$k\|D^2U\|_{L^2(D_a)}\|\nabla\chi_\varepsilon\|_{L^2(D_a)} \longrightarrow 0.$$

It follows first that $\int DX : DY_k = A\int X\cdot Y_k$ for every fixed $k$. Only now let $k\to\infty$, using the strong $H^1$ convergence of $Y_k$ proved above. This establishes (7.39) with no residual contribution from the ridge or vertex.

The positive- and negative-part chain rules, including the azimuthal derivative of $e_\rho$, give

$$DX : DY = |\nabla v|^2 + |\nabla w|^2 + \frac{v^2}{\rho^2}, \qquad X\cdot Y = v^2 + w^2 \quad \text{almost everywhere.}$$

Both scalars $v, w$ are admissible for the mixed Rayleigh minimum $A$, since their base traces vanish. Consequently (7.39) reads

$$\left(\int_{D_a} |\nabla v|^2 - A\int_{D_a} v^2\right) + \left(\int_{D_a} |\nabla w|^2 - A\int_{D_a} w^2\right) + \int_{D_a}\frac{v^2}{\rho^2} = 0. \tag{7.40}$$

All three terms are nonnegative, so $v = 0$ and $\int|\nabla w|^2 = A\int w^2$. The side relation $w = av$ then gives zero side trace as well, hence $w \in H^1_0(D_a)$. The full Dirichlet eigenvalue $\lambda_1^D(D_a)$ is strictly larger than $A$: equality in the variational inclusion would make a positive Dirichlet ground state a mixed minimizer, with both zero value and zero normal derivative on an open smooth side patch, contradicting the Hopf lemma. The Dirichlet Rayleigh inequality therefore forces $w = 0$. We have proved $U_\rho \le 0$ and $U_z \ge 0$ directly.

We finish the directional estimate, also recording its axis justification. Put $q = -s \geq 0$, $H = p - aq$, and $h = H_-$. Then $H \in H^1(D_a)$, its side trace is zero, and its base trace is nonnegative. Thus

$$h \in H_0^1(D_a), \qquad 0 \leq h \leq aq, \qquad h/\rho \in L^2(D_a).$$

Differentiating the cylindrical equation away from the axis gives

$$(-\Delta - A)H = \frac{aq}{\rho^2}.$$

Test with $\eta_\delta h$, where $\eta_\delta = 0$ for $\rho \leq \delta$, $\eta_\delta = 1$ for $\rho \geq 2\delta$, and $|\nabla \eta_\delta| \leq C/\rho$ in between. The error tends to zero, since on $E_\delta = D_a \cap \{\delta < \rho < 2\delta\}$ it is bounded by

$$C\|\nabla H\|_{L^2(E_\delta)}\|h/\rho\|_{L^2(E_\delta)}.$$

Also $qh/\rho^2 \leq aq^2/\rho^2 \in L^1$, so dominated convergence gives

$$\int_{D_a} |\nabla h|^2 + a\int_{D_a} \frac{qh}{\rho^2} = A\int_{D_a} h^2.$$

The same strict Dirichlet gap forces $h = 0$. Therefore $U_z \geq a|U_\rho|$. Squaring, integrating over the two half-cones, and using $A = \int_{\mathcal{C}_a}(|U_z|^2 + |U_\rho|^2)$ proves the energy estimate in (7.33) as well. □

**Lemma 7.7** (A strict downward crossing at every coincidence)**.** *For $a > 0$, the functions*

$$A(a) = \lambda_\parallel(\mathcal{C}_{a,1}), \qquad T(a) = \lambda_\perp(\mathcal{C}_{a,1})$$

*are continuously differentiable. At every $a$ such that $A(a) = T(a)$, they satisfy*

$$(A - T)'(a) < -\frac{1}{6a}. \tag{7.41}$$

*Proof.* We first prove a transverse slice estimate using only polynomials. Let $\gamma$ be the first mixed eigenvalue on the half unit disk $B_+ = \{y \in \mathbb{R}^2 : |y| < 1,\ y_1 > 0\}$, with Dirichlet condition on its diameter and Neumann condition on its semicircle. The polynomial

$$P(y) = y_1\left(1 - \frac{3}{8}|y|^2 + \frac{1}{40}|y|^4\right)$$

is positive in $B_+$, vanishes on the diameter, and has zero outward normal derivative on the semicircle. Moreover,

$$-\Delta P - 3P = \frac{y_1|y|^2(21 - 3|y|^2)}{40} > 0 \qquad \text{in } B_+.$$

Green's formula against its positive mixed ground state $\phi$ gives $(\gamma - 3)\int_{B_+} P\phi > 0$, so $\gamma > 3$. Odd reflection, the variational principle, and scaling show that every $H^1$ function odd in $y_1$ on a disk of radius $r$ satisfies

$$\int |\nabla_y v|^2 \geq \frac{\gamma}{r^2}\int v^2.$$

For an $L^2$-normalized transverse odd ground state $V$ on $\mathcal{C}_{a,1}$, apply this inequality to almost every horizontal slice, whose radius is $r(z) = 1 - |z|/a \leq 1$. Slicing and integration yield

$$E_y^T := \int_{\mathcal{C}_{a,1}} |\nabla_y V|^2 \geq \gamma\int_{\mathcal{C}_{a,1}} V^2 > 3. \tag{7.42}$$

This argument applies to the whole transverse odd sector and does not require an angular-mode classification.

We next justify differentiability despite the moving cone edges. The unitary pullback $f(y,t) = \sqrt{a}\,W(y,at)$ sends each odd sector to its fixed closed subspace of $H^1(\mathcal{C}_{1,1})$. Its Rayleigh form is

$$b_a(f,g) = \int_{\mathcal{C}_{1,1}} \left(\nabla_y f \cdot \nabla_y g + a^{-2} f_t g_t\right), \tag{7.43}$$

with fixed $L^2$ inner product. The sector ground state is simple by the mixed-ground-state argument in the preceding proof. For the transverse sector this means simplicity on the fixed $y_1$-odd subspace; other transverse directions may give the same full Neumann eigenvalue, but do not give additional ground states in this subspace. Consider the constrained equation

$$b_a(f,\cdot) = \lambda\langle f,\cdot\rangle, \qquad \|f\|_2 = 1.$$

Its linearization in $(f,\lambda)$ is invertible. On the $L^2$-orthogonal complement of the ground state, the spectral gap gives, writing $\lambda_+ > \lambda$ for the next sector eigenvalue,

$$b_a(w,w) - \lambda\|w\|_2^2 \geq \left(1 - \frac{\lambda}{\lambda_+}\right) b_a(w,w).$$

The odd Poincaré inequality and the positive form coefficients make this coercive in $H^1$. In the ground-state direction, the normalization and multiplier provide the remaining equations. Lax–Milgram and the implicit function theorem therefore give a locally continuously differentiable normalized eigenpair.

To identify this branch with the sector minimum, first note that comparison of the forms at $a$ and $a_0$ gives

$$\min\{1, a_0^2/a^2\}\lambda(a_0) \leq \lambda(a) \leq \max\{1, a_0^2/a^2\}\lambda(a_0).$$

Thus the minimum is continuous without any differentiability assumption. If $a_j \to a_0$, its normalized grounds, positive on the appropriate half-domain, are bounded in $H^1$ on the fixed cone. Weak $H^1$ and strong $L^2$ compactness, form comparison, and simplicity imply that every subsequential limit is the same normalized ground state at $a_0$. Equality of the limiting form energies then upgrades the convergence to strong $H^1$. These eigenpairs consequently belong to the implicit-function neighborhood and equal its unique branch. The minima themselves are therefore continuously differentiable.

Differentiating the equation and the normalization gives the Hellmann–Feynman identities. In physical coordinates they are

$$A'(a) = -\frac{2}{a}E_z^A, \qquad T'(a) = -\frac{2}{a}E_z^T, \qquad E_z^A = \int_{\mathcal{C}_{a,1}} |U_z|^2, \quad E_z^T = \int_{\mathcal{C}_{a,1}} |V_z|^2, \tag{7.44}$$

where both eigenfunctions have full-cone $L^2$ norm one. Indeed, differentiating (7.43) gives $-2a^{-3}\int |f_t|^2$, and the unitary pullback satisfies $\int |f_t|^2 = a^2 \int |W_z|^2$.

Suppose now that $A(a) = T(a) = \lambda$. The comparisons (7.21) and (7.22) give $a > \sqrt{5}/2 > 1$. The universal transverse upper bound (7.18) gives

$$\lambda < \frac{83}{15} + \frac{3}{10a^2} < \frac{35}{6}. \tag{7.45}$$

Lemma 7.6 and (7.42) imply

$$E_z^A \geq \frac{a^2}{1+a^2}\lambda, \qquad E_z^T = \lambda - E_y^T < \lambda - 3.$$

Consequently,

$$E_z^A - E_z^T > 3 - \frac{\lambda}{1+a^2} > 3 - \frac{35}{12} = \frac{1}{12}. \tag{7.46}$$

The identities (7.44) now yield (7.41). □

*Proof of Theorem 7.5.* Set $D = A - T$ as in Lemma 7.7. The comparisons (7.21) and (7.22) give

$$D(a) > 0 \quad \left(0 < a \leq \frac{\sqrt{5}}{2}\right), \qquad D(a) < 0 \quad \left(a \geq \sqrt{\frac{339}{181}}\right),$$

and at least one zero between these thresholds. By Lemma 7.7, $D$ is $C^1$ and every zero has strictly negative derivative.

At any zero $b$, differentiability gives $D(b+s) = D'(b)s+o(|s|)$. Thus $D$ is positive immediately to its left and negative immediately to its right. These local signs force global uniqueness as follows. If $b < c$ were two zeros, then $D < 0$ immediately to the right of $b$, while $D > 0$ immediately to the left of $c$. Choose $b < x < y < c$ with $D(x) < 0 < D(y)$ and let

$$d = \min\{t \in [x, y] : D(t) = 0\}.$$

The intermediate value theorem makes the zero set in $[x, y]$ nonempty, and continuity makes it compact, so its minimum $d$ exists. Moreover $x < d < y$ and $D < 0$ on $[x, d)$: a positive value before $d$ would give an earlier zero by continuity. For every $0 < s < d - x$ one therefore has

$$\frac{D(d) - D(d - s)}{s} > 0.$$

Letting $s \downarrow 0$ yields $D'(d) \geq 0$, contradicting the strictly negative derivative at every zero. This argument does not assume that the zero set is finite. The unique zero $a_*$ therefore satisfies (7.30), and (7.41) gives (7.32). Continuity and the absence of any other zero give $D > 0$ below $a_*$ and $D < 0$ above $a_*$.

Finally, $\mathcal{C}_{L,R} = R\mathcal{C}_{L/R,1}$, and uniform scaling multiplies both sector eigenvalues by $R^{-2}$. This proves (7.31) for every $L, R > 0$. □

Theorem 7.5 determines the ordering throughout the interval (7.23): its only equality ratio is $a_*$. In particular, the equality ratio is strictly larger than one, in contrast with the planar rhombus.

For comparison, the two branches also admit exact weighted variational descriptions. Scale to $R = 1$ and use the fixed triangle $T$ above. Rotational symmetry and positivity of the mixed ground states reduce the axial sector to angular order zero and the transverse sector to angular order one. For the latter, Fourier decomposition on the half-circle $y_1 > 0$ has first Dirichlet angular mode $\cos\theta$; every higher mode has a larger angular-energy coefficient. Hence

$$\lambda_{\parallel}(a) = \inf_{u|_{t=0}=0} \frac{\displaystyle\int_T s(u_s^2 + a^{-2}u_t^2)\, \mathrm{d}s\, \mathrm{d}t}{\displaystyle\int_T su^2\, \mathrm{d}s\, \mathrm{d}t}, \tag{7.47}$$

$$\lambda_{\perp}(a) = \inf_{v} \frac{\displaystyle\int_T \left[s(v_s^2 + a^{-2}v_t^2) + \frac{v^2}{s}\right] \mathrm{d}s\, \mathrm{d}t}{\displaystyle\int_T sv^2\, \mathrm{d}s\, \mathrm{d}t}. \tag{7.48}$$

The infima run over nonzero functions whose associated three-dimensional modes belong to $H^1$; in particular, the transverse angular energy $\int_T v^2/s$ is finite. The transverse problem has a natural Neumann condition at $t = 0$. The weight $s$ and the angular term $v^2/s$ prevent exchange of the triangle's two legs from giving the planar comparison. Theorem 7.5 nevertheless gives a single transition in (7.23) through a derivative estimate at every coincidence. The transition constant is characterized uniquely; no elementary closed formula for it is asserted.

7.3. **Quartic thresholds and localization of coincidences.** For the quartic bodies (1.9), write $a = L/R$. Their squared meridian profile is

$$r(z)^2 = R^2\left[1 - (1 - c)\frac{z^2}{L^2} - c\frac{z^4}{L^4}\right], \qquad 0 < c < 1. \tag{7.49}$$

**Corollary 7.8** (Quartic spectral thresholds)**.** *For every $n \geq 2$, the quartic family satisfies*

$$a \geq \sqrt{1+c} \quad \Longrightarrow \quad \lambda_{\parallel} < \lambda_{\perp}, \tag{7.50}$$

$$a \leq \sqrt{1-c} \quad \Longrightarrow \quad \lambda_{\parallel} > \lambda_{\perp}. \tag{7.51}$$

*For each fixed $c \in (0,1)$, there is a finite nonempty set of coincidence aspect ratios. Every such ratio lies in*

$$\sqrt{1-c} < a < \sqrt{1+c}, \tag{7.52}$$

*and every corresponding domain is a nonball.*

*Proof.* Proposition 6.4 verifies (G1)–(G3) and nonellipsoidality for these domains.

Direct differentiation gives

$$\mathcal{F}(z) = z + r(z)r'(z) = z\left[1 - \frac{1}{a^2}\left(1 - c + 2c\frac{z^2}{L^2}\right)\right]. \tag{7.53}$$

If $a \geq \sqrt{1+c}$, this expression is strictly positive for $0 < z < L$. If $a \leq \sqrt{1-c}$, it is strictly negative there. Theorem 1.3 proves (7.50) and (7.51), including both endpoint ratios. In particular, every coincidence satisfies (7.52).

With $c$ and $R$ fixed, varying $L$ gives an axial stretching family. Proposition 6.2 therefore gives a finite nonempty coincidence set. Since $c > 0$, the nonzero quartic coefficient in the squared profile prevents any member of this family from being a ball. □

*Remark* 7.9. For every ratio in (7.52), formula (7.53) is positive near $z = 0$ and negative near $z = L$. Nevertheless, both strict spectral orderings occur within this interval. Indeed, the strict gaps at its two endpoints persist by continuity under stretching, so moving either endpoint slightly into the interval gives the corresponding strict ordering with sign-changing $\mathcal{F}$. Thus a sign change of $\mathcal{F}$ does not imply a coincidence.

7.4. **A unique quartic coincidence near the ball.** The localization above gives a precise transition for small positive $c$.

**Theorem 7.10** (The quartic transition near the ball)**.** *Fix $n \geq 2$. There exist $\delta_n \in (0,1)$ and a real-analytic function $a_*(c)$ near $c = 0$, with $a_*(0) = 1$, such that for every $0 < c < \delta_n$ the quartic family has exactly one coincidence aspect ratio, namely $a = a_*(c)$. More precisely,*

$$\begin{aligned} &\lambda_{\parallel} > \lambda_{\perp} \quad \text{if } 0 < a < a_*(c), \\ &\lambda_{\parallel} = \lambda_{\perp} \quad \text{if } a = a_*(c), \\ &\lambda_{\parallel} < \lambda_{\perp} \quad \text{if } a > a_*(c). \end{aligned} \tag{7.54}$$

*The transition has the expansion*

$$a_*(c) = 1 - \frac{n-2}{2(n+4)}c + O(c^2) \qquad (c \to 0). \tag{7.55}$$

*For $n \geq 3$, one has $a_*(c) < 1$ for all sufficiently small positive $c$. In dimension two, the asserted expansion is $a_*(c) = 1 + O(c^2)$.*

*Proof.* Uniform scaling multiplies both eigenvalues by the same positive factor, so we set $R = 1$ and $a = L$. For $(a, c)$ near $(1, 0)$, consider

$$T_{a,c}(y, z) = (\sqrt{1 + cz^2}\, y, az), \qquad (y, z) \in \overline{B^n}. \tag{7.56}$$

These are analytic diffeomorphisms onto their images. For $c \geq 0$ sufficiently small, the image is exactly the quartic body: on each slice, $|y|^2 < 1 - z^2$ becomes

$$|\widetilde{y}|^2 < (1 - z^2)(1 + cz^2), \qquad \widetilde{z} = az.$$

For small negative $c$, use the same images $T_{a,c}(B^n)$ to extend the family. They are smooth strictly convex deformations of the ball. The maps preserve all coordinate reflections.

Let $j_{a,c} = \det DT_{a,c}$. Pullback gives the energy and mass forms on $H^1(B^n)$,

$$q_{a,c}(f,g) = \int_{B^n} j_{a,c}(DT_{a,c}^{-T}\nabla f)\cdot(DT_{a,c}^{-T}\nabla g),$$
$$m_{a,c}(f,g) = \int_{B^n} j_{a,c} fg.$$

Their coefficients depend analytically on $(a,c)$, the energy coefficients are uniformly elliptic near $(1,0)$, and the mass density is positive. Reflection preservation fixes the two form domains: functions odd in $z$ and functions odd in $y_1$, respectively.

In each of these odd subspaces the lowest eigenvalue at the ball is simple, by Lemma 3.1. The normalized lowest eigenpair consequently depends analytically on $(a,c)$. Here is the local invertibility argument. Let $A_{a,c}$ and $B_{a,c}$ denote the operators into the dual of the fixed odd subspace represented by $q_{a,c}$ and $m_{a,c}$. Apply the analytic implicit function theorem to

$$(u,\lambda,a,c) \longmapsto \left(A_{a,c}u - \lambda B_{a,c}u,\ \frac{m_{a,c}(u,u)-1}{2}\right).$$

At the ball, decompose $u$ into the normalized ground-state line and its $L^2$-orthogonal complement. Testing the linearized eigenvalue equation against the ground state determines the eigenvalue variation, and the normalization determines the component along that line. If $\lambda_+$ is the next eigenvalue in the odd subspace and $\lambda = \beta_n^2$ is its lowest one, then on the complement

$$q_{1,0}(w,w) - \lambda\|w\|_2^2 \geq \left(1 - \frac{\lambda}{\lambda_+}\right) q_{1,0}(w,w).$$

The odd Poincaré inequality makes this coercive in $H^1$, so Lax–Milgram gives the remaining inverse. Compactness and min–max continuity identify the analytic local branch with the lowest eigenvalue in that subspace. Thus

$$D(a,c) = \lambda_{\|}(a,c) - \lambda_{\perp}(a,c) \tag{7.57}$$

is analytic near $(1,0)$, and $D(1,0) = 0$.

We compute its first derivatives directly from the forms. Let $x \mapsto x + \varepsilon Z(x) + O(\varepsilon^2)$ be a smooth deformation of the ball that preserves the relevant reflections, and let $u$ be an $L^2$-normalized eigenfunction in a simple odd sector. Differentiating the weak eigenpair equation and its normalization cancels the terms involving the derivative of $u$ and gives

$$\lambda'(0) = \int_{B^n} \left[(\operatorname{div} Z)(|\nabla u|^2 - \lambda u^2) - 2\nabla u \cdot DZ\nabla u\right]. \tag{7.58}$$

Since $\Delta u = -\lambda u$, the integrand is the divergence of

$$(|\nabla u|^2 - \lambda u^2)Z - 2(Z\cdot\nabla u)\nabla u.$$

The Neumann condition therefore yields

$$\lambda'(0) = \int_{S^{n-1}} (|\nabla_\tau u|^2 - \lambda u^2) h \, \mathrm{d}S, \qquad h = Z\cdot\omega. \tag{7.59}$$

This derivation takes place in a simple odd sector, so the multiplicity of the full ball eigenvalue causes no ambiguity.

Write $\omega = (\omega_1,\dots,\omega_{n-1},\omega_z)$. The normalized axial and transverse ball modes have boundary values $b\omega_z$ and $b\omega_1$, with the same $b > 0$. To see this within the present framework, Lemma 2.4 and Corollary 6.1 make $J : E_1(B^n) \to \mathbb{R}^n$ an equivariant bijection. Choosing $J(u_i) = e_i$, equivariance forces $(u_1(x),\dots,u_n(x)) = g(|x|)x$; the coordinate modes have a common $L^2$ norm, so their common normalization gives the stated boundary traces. Their positive signs on the corresponding

half-balls and Hopf's lemma with the Neumann condition give $b > 0$. Since $|\nabla_S \omega_i|^2 = 1 - \omega_i^2$, subtraction in (7.59), with $K = b^2(1+\lambda) > 0$, gives

$$D'(h) = -K \int_{S^{n-1}} h(\omega)(\omega_z^2 - \omega_1^2)\, \mathrm{d}S. \tag{7.60}$$

The map (7.56) supplies the normal velocities

$$h_a = \omega_z^2, \qquad h_c = \frac{1}{2}\omega_z^2(1-\omega_z^2).$$

For normalized spherical averages $\mathbb{E}$, the required moments are

$$\begin{aligned} \mathbb{E}\omega_z^4 &= \frac{3}{n(n+2)}, & \mathbb{E}(\omega_z^2\omega_1^2) &= \frac{1}{n(n+2)}, \\ \mathbb{E}\omega_z^6 &= \frac{15}{n(n+2)(n+4)}, & \mathbb{E}(\omega_z^4\omega_1^2) &= \frac{3}{n(n+2)(n+4)}. \end{aligned} \tag{7.61}$$

For example, these follow by integrating the corresponding Gaussian monomials in polar coordinates. Substitution gives

$$\begin{aligned} D_a(1,0) &= -\frac{2K|S^{n-1}|}{n(n+2)} < 0, \\ D_c(1,0) &= -\frac{K|S^{n-1}|(n-2)}{n(n+2)(n+4)}. \end{aligned} \tag{7.62}$$

The analytic implicit function theorem now gives a unique local zero graph $a = a_*(c)$, and

$$a_*'(0) = -\frac{D_c(1,0)}{D_a(1,0)} = -\frac{n-2}{2(n+4)}.$$

This proves (7.55).

It remains to exclude coincidences outside this local neighborhood. Corollary 7.8 confines every coincidence for $0 < c < 1$ to $(\sqrt{1-c}, \sqrt{1+c})$. This interval contracts to 1 as $c \downarrow 0$. After reducing $\delta_n$, every coincidence therefore lies in the fixed implicit-function neighborhood. There is exactly one coincidence for all $a > 0$. Continuity under stretching and the two strict endpoint orderings in Corollary 7.8 give (7.54). The dimensional conclusions follow from the linear coefficient in (7.55). □

*Remark* 7.11. The constant $\delta_n$ is not quantified, and (7.55) is an asymptotic formula rather than an explicit expression for the transition. For $n = 2$, its vanishing linear coefficient does not assert that $a_*(c)$ is identically one. For $n \geq 3$ and sufficiently small positive $c$, any $a \in (a_*(c), 1)$ gives $L < R$ together with $\lambda_{\|} < \lambda_{\perp}$. Thus the ordering cannot in general be determined by comparing $L$ and $R$ alone.

## References


[1] H. Chen, Y. Li, and L. Wang, *Monotone properties of the eigenfunction of Neumann problems*, J. Math. Pures Appl. (9) **130** (2019), 112–129. doi:10.1016/j.matpur.2019.01.013.

[2] J. de Dios Pont, *Convex sets can have interior hot spots*, preprint, 2024. arXiv:2412.06344.

[3] D. Jerison and N. Nadirashvili, *The "hot spots" conjecture for domains with two axes of symmetry*, J. Amer. Math. Soc. **13** (2000), no. 4, 741–772. doi:10.1090/S0894-0347-00-00346-5.

[4] J. B. Kennedy and J. Rohleder, *On the hot spots conjecture in higher dimensions*, Trans. Amer. Math. Soc. Ser. B **13** (2026), 108–131. doi:10.1090/btran/245. Preprint: arXiv:2410.00816.

[5] B. Klartag, *A Berry–Esseen type inequality for convex bodies with an unconditional basis*, Probab. Theory Related Fields **145** (2009), nos. 1–2, 1–33. doi:10.1007/s00440-008-0158-6. Preprint: arXiv:0705.0832.

[6] M. Mitrea, *Dirichlet integrals and Gaffney–Friedrichs inequalities in convex domains*, Forum Math. **13** (2001), no. 4, 531–567. doi:10.1515/form.2001.021.

[7] B. Siudeja, *On mixed Dirichlet–Neumann eigenvalues of triangles*, preprint, 2015. arXiv:1501.07618v1.